

A COMPLETE REFERENCE

The Caffevolve™ Platform

THE PLATFORM	CAFFEINE MANAGEMENT	ADAPTIVE FEEDBACK	THE USER PROFILE	THE DEVICE	CAFFEVELOVEGO	USER EXPERIENCE
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Confidential

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1. The Caffevolve™ Platform

Caffevolve™ is a comprehensive caffeine management platform built to provide coffee consumers with timely, precise guidance on their caffeine intake from the moment they wake to the moment they retire to bed. The primary objective of this guidance is to keep the consumer within a caffeine range that keeps them alert and focused when they need to be, without crossing over that fine line that typically results in agitation, nervousness, and ultimately, a night of poor sleep. There is a science to this objective and the Caffevolve™ Platform has captured it precisely.

The Caffevolve™ Shift

Caffeine management doesn't place a ceiling on your cups. It turns your caffeine into a budget you spend over the course of the day.

For most of coffee's history, every cup has been a blunt yes-or-no gamble — drink it and risk feeling wired and on edge now, or losing sleep later; skip it and go without the focus and pleasure. Caffevolve™ ends that trade-off by treating caffeine not as a fixed property of the cup, but as a dial. Because the device blends caffeinated and decaffeinated beans to order, your daily caffeine stops being a limit on how many cups you can enjoy and becomes a budget you can spend as you like — diluted across more cups, stretched further into the day and evening, while keeping you clear of the jittery overstimulation of too much and gentle on your sleep at night. A 7 p.m. cup is no longer off-limits; it's simply composed so it won't leave you wired or keep you up. That is the Caffevolve™ Shift: the warmth, the ritual, the taste you reach for — without the agitation or the lost sleep that used to come with it. Not less coffee. More — on your terms.

At the heart of the platform is a simple but powerful idea: not everyone responds to caffeine the same way, and the same person responds differently depending on how much caffeine is already in their system. Caffevolve™ accounts for both. Every brew recommendation begins with the user's personal caffeine sensitivity — their individual tolerance level — and combines it with a real-time picture of how much caffeine is still active in their body from prior consumption. The goal is to keep the user in their personal Caffeine Comfort Zone: alert, focused, and performing well — above the level where caffeine has a muted impact, and below the level where too much of it starts working against them. Get it right, and caffeine becomes a precision tool rather than a daily gamble.

Caffevolve™ is always thinking ahead. The platform continuously tracks how the caffeine present in your system now will “decay” in the hours ahead and projects forward to your declared bedtime — so it knows whether a brew right now will clear your system enough to let you sleep well. If it won't, the caffeine recommendation is adjusted

to keep that from happening. The result is a system that manages caffeine as a full-day variable, not a cup-by-cup decision — one that works for you around the clock, not just in the moment you press brew.

The Caffevolve™ Device

The Caffevolve™ device is the heart of the platform — and what makes it unlike anything else is that it doesn't just calculate and recommend your ideal caffeine level — it brews it. Using a dual-hopper system that blends caffeinated and decaffeinated beans in the exact ratio the caffeine blend recommendation calls for, the device closes the loop between calculation and cup. But how the platform arrives at that recommendation — and how much control the user retains over it — is just as deliberate as the brew itself.

The Caffevolve™ Advisory Philosophy

The system recommends. The user decides. The machine always brews.

The caffeine-blend recommendation is always the starting point — never a restriction. The user can accept it, adjust it, or override it entirely — a Manual Caffeine Adjustment that the platform honors without question. Caffevolve™ is an intelligent advisor, not a gatekeeper. What ends up in your mug is always your call.

The Numerical Rationale Principle

Specific published figures cited in this document (e.g., caffeine safety thresholds, half-life ranges, extraction efficiencies) are reference points used to establish the scientific rationale for the platform's methodology — they are not themselves the invention. The protectable core is the process by which the system derives, personalizes, and continuously refines these values from individual user behavior, independent of which literature-sourced constants happen to seed it. Specific citations are maintained separately and should be verified independently before use in patent claim language or regulatory submissions.

The Caffevolve™ device is home base. It's where most of your daily caffeine is brewed and consumed. The device logs and tracks the caffeine addition of each new brew while recognizing how much of the caffeine clears from your system (decays) as the hours pass. It's always calculating where you currently stand and projecting where you'll be at key moments later in the day.

But its home-based reach has a limit.

The moment you step away and consume caffeine elsewhere — a coffee shop, a convenience store — the chain is broken. What the device doesn't know, it can't account for. The calculations drift. The recommendations that follow are built on an incomplete picture. This is a real problem. This is where the companion app steps in.

Caffeolve™ Go

The Caffeolve™ companion app is the platform's away-from-home intelligence. When the user steps out — to a coffee shop, a meeting, a workout, anywhere — the app keeps the caffeine picture complete. It does this in two ways. First, it lets the user log any caffeine consumed away from the device — a quick entry that immediately updates the platform's running calculation, as if the device had brewed it. Second, it advises. Before the user orders, the app looks at their current caffeine level and the time remaining before bedtime, and recommends what still fits. When the user walks out the door, the app keeps the chain intact.

Together, the device and the Caffeolve Go app form a closed loop — one that follows the user through their entire day, from the first brew of the morning to the last decision of the evening. The Caffeolve™ Platform doesn't manage coffee. It manages caffeine — intelligently, continuously, and always in service of the user's performance and rest. And because the platform works entirely from the user's own behavioral signals and declared bedtime, it is fully schedule-agnostic: **it works equally well for a nurse on rotating shifts, a third-shift parts assembler, a student, or a traditional nine-to-five professional.**

To do this well, the platform is built on a working model of how caffeine behaves in the human body — how it accumulates, how it clears as the hours pass, and how individual caffeine sensitivity shapes the experience. The next section covers exactly that.

2. The Caffevolve™ Approach to Caffeine Management

At the center of the Caffevolve™ approach to caffeine management is the development of a user's personal Caffeine Comfort Zone — the caffeine band where they are alert, focused, and performing well, above the level where caffeine has a muted impact and below the level where it starts working against them. Three boundaries define this band: the Peak Zone Floor, the Wake Hours Caffeine Ceiling, and the Bedtime Caffeine Ceiling — and in Caffevolve™, all three are derived from the user's own reported behavior.

A. Building the Caffeine Comfort Zone

Caffeine Comfort Zone — Defined

Throughout this document we refer to the user's "Caffeine Comfort Zone" — the band where caffeine is actively working for them.

Caffevolve™ also calls this the Peak Zone. The Peak Zone sits between three boundaries: the Peak Zone Floor at the bottom, the Wake Hours Caffeine Ceiling at the top, and the Bedtime Caffeine Ceiling between them. The two ceilings play different roles: the Wake Hours Ceiling governs the active hours of the day, capping how much caffeine the user can carry at any moment before the focus and productivity benefit reverses. The Bedtime Ceiling governs the night, setting the maximum residual caffeine targeted in the user's system at the moment they go to sleep — and shaping every recommendation made during the hours that lead there. Within the Peak Zone, the caffeine present in the system delivers its intended benefit: alertness, focus, and sustained cognitive performance. Below it, caffeine in the system isn't enough to produce a meaningful effect. Above it, the benefit reverses — alertness gives way to agitation, focus to restlessness, and performance suffers, as does the ability to sleep.

How Caffevolve™ Learns You — The Caffeine Behavior Instrument (CBI)

The Caffeine Behavior Instrument is the platform's onboarding tool for establishing the initial boundaries of the user's Caffeine Comfort Zone and their personal Caffeine Decay Rate — the two inputs the platform needs before it can make a meaningful recommendation. Both are derived from the same place: the user's own reported behavior. How do they respond to caffeine over the course of the day? How quickly does it clear their system? The conventional approach to answering these questions is to ask the user to classify themselves: "Are you sensitive to caffeine?" The problem is that this question is nearly unanswerable.

The problem is that sensitivity, clearance, and tolerance are biologically independent traits that feel identical from the inside. A heavy drinker who reports "caffeine doesn't affect me" may be genuinely insensitive — or simply tolerant to a dose that would still wreck their sleep. Self-assessment can't tell the difference.

Caffeolve™ sidesteps this dilemma entirely. Instead of asking users to rate themselves on a trait they cannot accurately perceive, the platform asks about their own familiar behavior against known quantities — things a user can report reliably because they live them virtually every day. Some of these questions include:

- What kind of caffeinated drink do you enjoy first thing in the morning?
- How many cups does it take before you begin to feel alert, focused, and ready for the day?
- How many cups do you usually have before you feel wired, jittery, or overstimulated?
- What is the latest time of day you can have caffeine without it having a negative impact on your sleep?

From this initial round of questions, the platform begins to compute the user’s three Caffeine Comfort Zone boundaries and the user’s personal Caffeine Decay Rate (Section 3).

No sensitivity tier is assigned. No category is selected. The boundaries themselves are the quantified expression of the user’s Caffeine Sensitivity Level — derived from actual user behavior rather than self-assessed. Descriptive labels used elsewhere in this document — such as “moderate sensitivity” — characterize a computed result for the reader; they are not tiers, and no such category is selected or stored for Resident Users. (Guest profiles are the sole exception, where Higher / Moderate / Lower is a deliberately selected conservatism shortcut.)

These boundaries are starting points — designed to be right enough to guide the first brew and conservative enough to do no harm. From there, the platform’s Adaptive Feedback System (described in Section 3) refines them from real outcomes, so the Caffeine Comfort Zone continues to sharpen as the system gathers real-world user feedback.

SU — PROFILE STEP 2 OF 4

Tell Us About Your Daily Coffee Routine

4 / 8

WHEN IT'S TOO MUCH

Think of a day you'd had too much — jittery, wired, heart racing, hard to settle. About how many had you had by then?

Around two

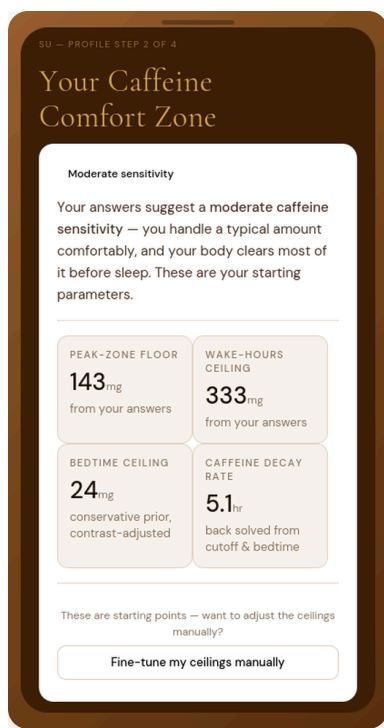
Three

Four

Five or more

I never really hit that point fallback

The Caffeine Behavior Instrument — question 04/08. The user describes a day they felt overstimulated. The platform back-calculates the Wake Hours Ceiling from the reported cup count.



Moderate Caffeine Sensitivity Level — the initial, computed Caffeine Comfort Zone and the Caffeine Decay Rate derived from the responses to the eight CBI questions.

Caffeine Comfort Zone — Establishing the Boundaries

Peak Zone Floor

The Peak Zone Floor is the minimum residual caffeine level at which caffeine is actively working for a given user — below it, the caffeine in their system isn’t producing a meaningful alertness benefit, and the user may feel sluggish, unfocused, or simply flat. The platform derives this from the user’s answer to a straightforward question: how many cups of their typical morning drink does it take before they feel alert and focused? The answer, paired with the known caffeine content of that drink, yields the residual level at which their body crosses from “caffeine present but not yet working” to “caffeine engaged.” This is their floor — the lower boundary of their Peak Zone.

Wake Hours Caffeine Ceiling

The Wake Hours Caffeine Ceiling is the upper boundary of the Caffeine Comfort Zone during the user’s active hours — the line above which caffeine stops being productive and starts being counterproductive. Alertness becomes agitation; focus becomes scattered attention; the user is overcaffeinated and, paradoxically, less productive than they would have been at a lower level. The platform derives this from the same onboarding sequence: how many cups of that same morning drink before the user feels overstimulated? The answer, against the same known caffeine content, yields the personal level at which the benefit reverses. This ceiling holds steady throughout active hours and applies regardless of when those hours fall — making it fully schedule-agnostic.

The Wake Hours Ceiling is constant — it does not change based on time of day. It is the overstimulation threshold at all times. It only comes under meaningful pressure during waking hours, when caffeine is being consumed and accumulating; during sleep, caffeine decays and the ceiling is never challenged.

Bedtime Caffeine Ceiling

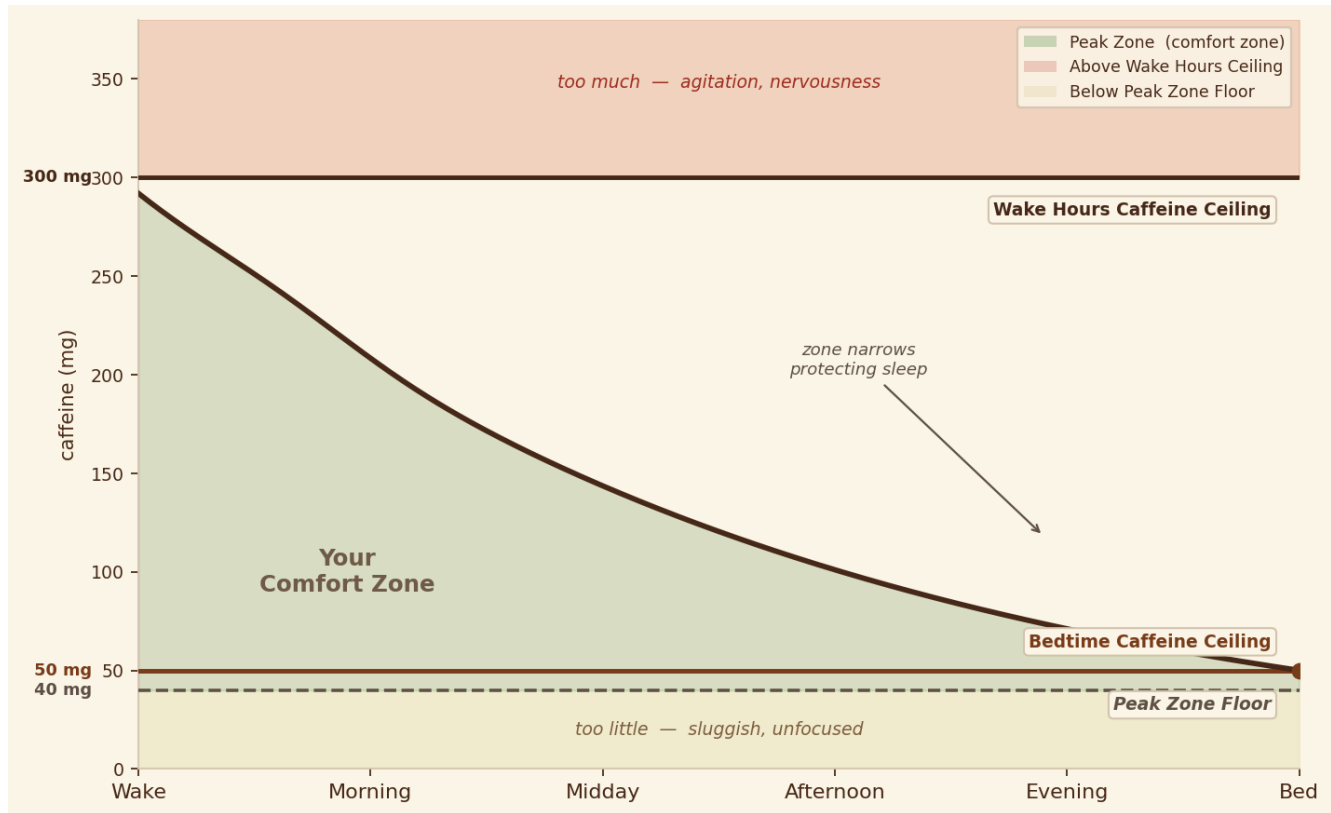
The Bedtime Caffeine Ceiling is the second of the platform's two ceilings — and the one that increasingly shapes recommendations as the day progresses and bedtime draws closer. Where the Wake Hours Ceiling is an instantaneous check against overstimulation right now, the Bedtime Ceiling is a forward-looking target: the maximum residual caffeine the platform targets at the moment the user goes to sleep. Every caffeine blend the platform recommends, from the first morning cup onward, is calibrated to ensure this target is honored — because a disrupted night of sleep is the single costliest failure the platform can produce.

The platform sets this boundary conservatively at onboarding — erring on the side of protecting sleep — and refines it from how the user actually sleeps. Because cutoff time informs both the decay rate and this ceiling, the platform weights the ceiling conservatively and lets the AFS separate the two signals as reported outcomes accumulate. Within days, the conservative starting point gives way to a boundary shaped by real outcomes.

Connecting the Dots

Together, these three boundaries define the user's personal Caffeine Comfort Zone — and the Caffeine Comfort Zone is the first component of the equation that drives every caffeine recommendation the platform makes. The other component is based on the user's Caffeine Decay Rate.

Defining Your Caffeine Comfort Zone



B. Determining Position Within the Caffeine Comfort Zone

The Caffeine Decay Rate (CDR)

The Caffeine Decay Rate is the personalized rate at which a user's body clears caffeine from their system — derived from their reported evening cutoff, as described in Section 2. It is the engine behind every time-based projection the platform makes: projected residual at the moment, projected residual at any hour, and projected residual at bedtime.

The underlying model is standard pharmacokinetics — exponential decay, in which the body clears a fixed proportion of its active caffeine in each unit of time. Peer-reviewed literature puts the individual half-life range at roughly 2.5 to 10 hours, with a population mean around five to six hours — which is where the platform begins before the user's personal rate is established. For every six hours that pass, approximately half of the user's active caffeine clears. A cup of coffee delivering 95 mg at 7:00 AM leaves roughly 47 mg by 1:00 PM, about 24 mg by

7:00 PM, and approximately 12 mg by 1:00 AM. Each cup follows this curve independently, and the platform sums them to produce the user's total active caffeine at any moment.

Because caffeine clearance is biologically independent of caffeine sensitivity, the decay rate cannot be inferred from the answers that set the Caffeine Comfort Zone boundaries — it must be isolated from a separate behavioral signal. The evening cutoff question provides exactly that, yielding a personal decay rate the platform applies to every subsequent projection. Because the Caffeine Decay Rate underlies every projection the platform makes — residual level, caffeine headroom, bedtime estimate — its accuracy has a compounding effect on the quality of every recommendation. The Adaptive Feedback System treats CDR refinement as its highest early priority, gathering decay-relevant signals as quickly as possible in the first weeks of use so that the platform's projections converge on the user's actual biology without delay.

Residual Caffeine Level

The Residual Caffeine Level is the total caffeine active in the user's system at any given moment — the computed output of applying the Caffeine Decay Rate to everything the user has consumed.

Accumulated from all brews and any externally logged caffeine entries, and continuously decaying on a 24-hour clock, it operates as a cumulative 24-hour model: not a stored value but a living projection that changes with every passing minute and every new brew.

Think of it as a running tally that moves in both directions. Every new brew or logged drink pushes it up; every hour that passes pulls it back down as the Caffeine Decay Rate works. At any point, the platform computes this number and compares it against the Caffeine Comfort Zone boundaries from Section 2.

With the residual level known, the picture is complete. The boundaries tell the platform where the edges are; the Caffeine Decay Rate tells it how fast caffeine is clearing; and the Residual Caffeine Level tells it where the user stands right now. From here, every recommendation is arithmetic.

C. Determining Distance to the Ceilings

The Dual-Ceiling Architecture

A single ceiling would be simple to manage. Caffeolve™ works with two, and it evaluates both at once.

The Dual-Ceiling Architecture is the platform's core decision model: every brew recommendation must satisfy the Wake Hours Caffeine Ceiling and the Bedtime Caffeine Ceiling simultaneously. Neither ceiling operates in isolation, and neither is allowed to override the other.

The two ceilings ask different questions. The Wake Hours Caffeine Ceiling is an instantaneous check — it asks whether adding this brew's caffeine to the user's current residual would push them above their focused and productive range and into overstimulation. The Bedtime Caffeine Ceiling is a forward-looking check — it asks whether the user's total residual at bedtime, with this brew included, will clear the targeted caffeine ceiling and leave the door open for sleep. One governs the next hour; the other governs the night.

Because the two checks measure different things, they rarely constrain at the same time. Early in the day, with a distant bedtime, the Wake Hours Caffeine Ceiling usually does the limiting — there are plenty of hours for caffeine to clear, so bedtime is not yet the constraint. Late in the day the situation reverses: the user may sit well within their wake-hours range, but with bedtime only a few hours off, the Bedtime Caffeine Ceiling becomes the tighter of the two. The platform checks both ceilings, every time. If either would be violated by the brew being considered, the caffeine recommendation is adjusted until the user stays under both — the strongest brew that clears each ceiling simultaneously.

Caffeine Headroom

Caffeine Headroom is the “caffeine distance” between the user’s current residual caffeine level and the nearest ceiling — the room available before the next boundary is reached. For every brew, the platform computes headroom independently against each ceiling: how much additional caffeine can be added before the Wake Hours Ceiling is reached, and how much before the Bedtime Ceiling is reached when projected forward to the user’s declared bedtime. The two checks are different in nature — the Wake Hours check is instantaneous, the Bedtime check is forward-looking, projecting the brew’s residual forward to the declared bedtime using the user’s personal CDR. Whichever ceiling leaves the least room becomes the binding headroom for that brew. The recommendation that follows is then calculated to fill that headroom exactly — the highest caffeinated blend the architecture allows at that moment.

The platform computes caffeine headroom with two tests — one for each ceiling — and takes the smaller of the two results.

THE TWO-TEST CAFFEINE HEADROOM CALCULATION

TEST 1 — WAKE HOURS (INSTANTANEOUS)

Caffeine Headroom = Wake Hours Ceiling - Current Residual

TEST 2 — BEDTIME (DECAY-AWARE)

Caffeine Headroom = (Bedtime Ceiling - Projected Bedtime Residual (prior)) ÷ Decay Factor

Decay Factor = $0.5^{(\text{Hours Until Bedtime} \div \text{Caffeine Decay Rate})}$

Projected Bedtime Residual (prior): the residual projected to bedtime from prior brews and logged drinks only — this brew not yet included.

Binding Caffeine Headroom = the lower result of Test 1 and Test 2

The first test checks the Wake Hours Ceiling directly. The second is decay-aware: it projects forward to bedtime using the Caffeine Decay Rate. The further off bedtime is, the more caffeine can be added now and still decay away in time — which is why the same residual level yields generous caffeine headroom at 8 AM and almost none at 9 PM. The platform runs both tests and adopts the smaller answer. That smaller number is the binding caffeine headroom — the true ceiling on what the platform will recommend.

Caffeine Headroom is the platform's internal working figure; the user never sees a milligram-of-headroom number on the device. What they see is the recommendation it produces — which is where the platform turns next.

Projected Residual at Bedtime

Of every number the platform computes, one matters more than the rest: the projected bedtime caffeine residual level. It is the answer to the question that defines the entire platform — if the user brews this now, where will they stand when they try to sleep?

D. Building the Caffeine Recommendation

The Candidate Brew

Caffeine Headroom tells the platform how much caffeine it has to work with. The next step is turning that allowance into a Candidate Brew — and the answer is a direct calculation, not a search.

The platform takes the binding caffeine headroom as its target caffeine output and applies the Blend Ratio Formula — factoring in the user's serving size, serving count, and brew strength — to compute the exact caffeinated-to-decaffeinated ratio that delivers that amount. One formula, one answer: the strongest brew that both ceilings allow. The result is the Candidate Brew — the fully-specified brew the platform has determined best serves the user at this moment, expressed as a blend ratio ready to be confirmed and brewed. That ratio becomes the Caffeine Blend Recommendation.

The Caffeine Blend Recommendation

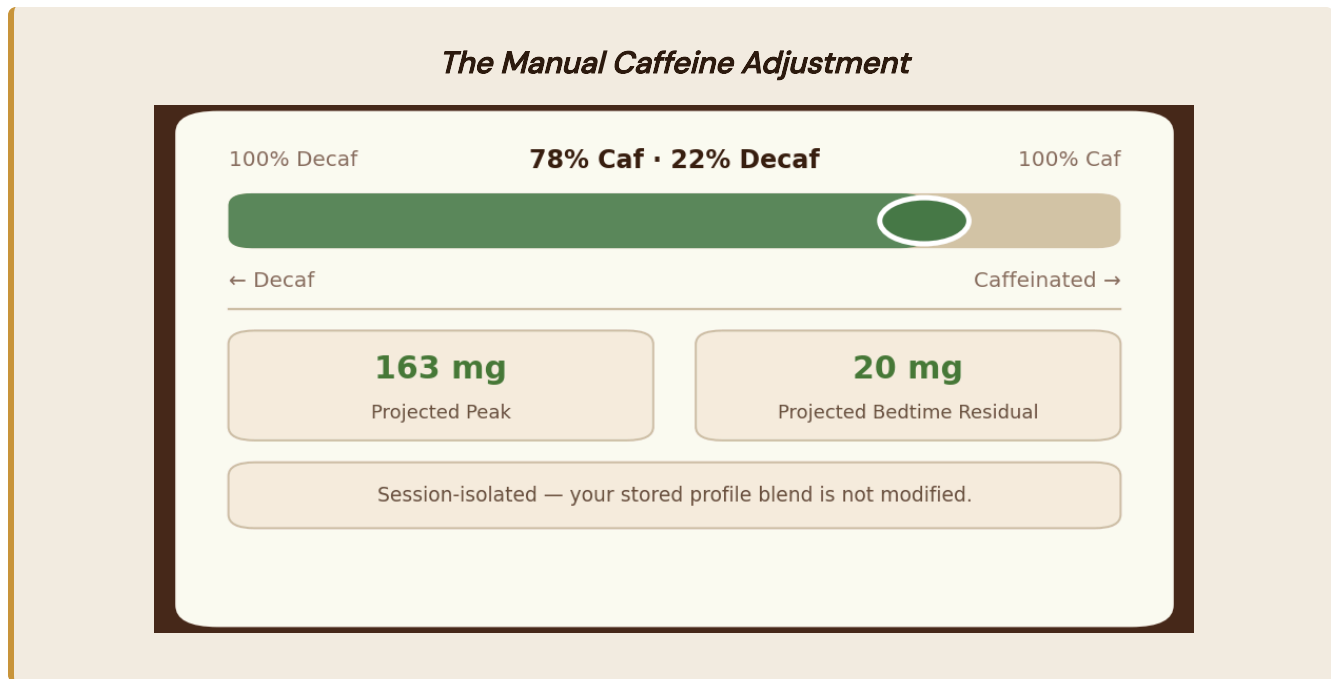
The Caffeine Blend Recommendation is the Candidate Brew presented to the user — the platform's complete answer translated into a form the user can read, understand, and act on. It is expressed as a blend ratio — the proportion of caffeinated to decaffeinated beans the device will grind together for this cup — because the ratio is the lever the user controls. A recommendation might read 70% caffeinated / 30% decaffeinated. That number tells the user exactly where they are, and exactly which direction to move if they want something different.

Alongside the ratio, the platform shows the user the two numbers that explain it: their post-brew caffeine residual level — where they'll stand after this cup — and the projected caffeine residual at bedtime — where this brew will leave them when it counts. The user sees not only what the platform recommends, but the reasoning behind it.

And then the platform stops recommending. The recommendation is the starting point — never a restriction. The user can accept the blend as presented, or choose to adjust it. The machine always brews

whatever the user confirms.

If the user wants to adjust the recommendation, they are advanced to the Manual Caffeine Adjustment — a slider that lets them dial the caffeine level up or down from the recommended blend. As the slider moves, the Projected Caffeine Peak and the Projected Bedtime Caffeine Residual update automatically, giving the user a live numeric guide to the consequence of every position. Whatever blend they choose becomes the blend that is brewed — and the platform logs it, updating the caffeine picture from that point forward.



The platform computes the projected bedtime residual (post-brew) — the figure the user sees on the recommendation screen, distinct from the prior figure the headroom test consumes — by taking the user’s residual level the moment the contemplated brew lands (current residual plus the caffeine the brew adds) and decaying that total forward to the declared bedtime using the user’s personal Caffeine Decay Rate. The result is a single figure in milligrams: the caffeine that will still be active in the user’s system at the moment they lie down.

This is the number that turns caffeine management from a daytime concern into a full-day one. A brew can sit comfortably within the Wake Hours Caffeine Ceiling and still be the wrong brew — if its projected residual at bedtime sits above the Bedtime Caffeine Ceiling, it will cost the user sleep. By projecting every candidate forward to bedtime before recommending it, the platform protects the night while it serves the day — or vice versa for a shift worker. It is the reason a Caffeolve™ recommendation is never just about the next cup — it is about how that cup lands hours later, when the lights go out.

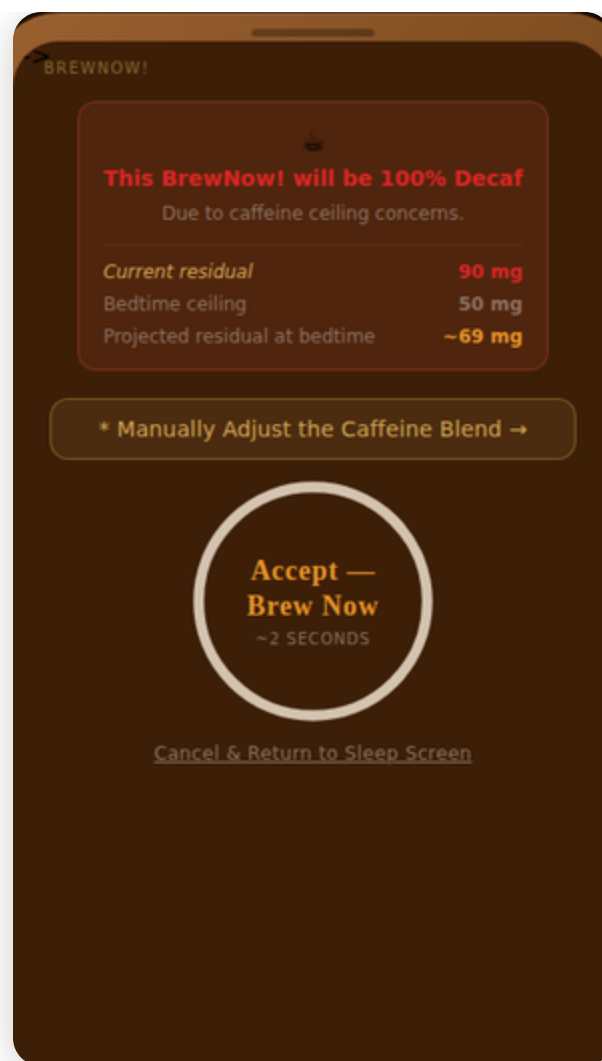
The Zero-Caffeine-Headroom State

There is one situation the platform must handle with particular care: the moment when no caffeinated brew fits and a new brew is requested. It happens when the user’s residual level — from earlier brews, logged drinks, or both — has already climbed so high, or bedtime sits so close, that even the smallest caffeinated candidate would

breach the binding ceiling. The caffeine headroom has gone to zero. The platform calls this the zero-caffeine-headroom state.

When no caffeinated candidate can satisfy both ceilings, the platform does not quietly recommend a weaker blend and hope for the best. It recommends a full 100% decaffeinated brew — and, just as importantly, it explains why. The user is told plainly that their current caffeine level, or the nearness of their bedtime, leaves no room for more caffeine right now, and that a decaf brew lets them enjoy the cup without putting the later hours of the day — or the night ahead — at risk.

This is the Advisory Philosophy at its most honest. Even here the platform does not lock the user out; the recommendation is still a starting point, and the user remains free to adjust. But the zero-caffeine-headroom state is the clearest expression of what Caffevolve™ is for — not to push caffeine, and not to forbid it, but to tell the user the truth about where they stand and let them decide with their eyes open.



BN-Decaf — The platform's response to a zero-headroom BrewNow! request. Current residual 90 mg, bedtime ceiling 50 mg, projected residual at bedtime ~69 mg. The platform recommends 100% decaf and explains why — the user retains the option to manually adjust.

The Caffevolve™ Approach to Caffeine Management — Case Studies

The following three scenarios illustrate the Caffevolve™ approach in action — each showing a different aspect of the system working on behalf of a real user in a real moment. Wayne is six cups in and headed in the wrong direction — the platform keeps him sharp without cutting him off. Claudia has a bedtime ceiling that won't move and an afternoon slump that won't wait — the platform finds her the most it can honestly offer. Diane has a board presentation at nine and a bedtime that's closer than she'd like — the platform finds the strongest brew that protects both.

Wayne on Deadline

A heavy caffeine day collides with an overstimulation ceiling. The platform intervenes before the next cup makes things worse.

It's the back half of a long day and the deadline has gone from "this week" to "tonight." Wayne is six cups in — the last three poured barely an hour apart as the pressure climbed — his knee is bouncing under the desk, and he's just read the same paragraph three times without keeping any of it. He's wide awake, wired tight, and getting almost nothing done.

Here's the part nobody warns you about. The closer the deadline got, the faster Wayne reached for the coffee — more cups, shorter gaps, more fuel in the tank. But somewhere around cup four the coffee quietly stopped helping and started working against him. He isn't tired. He's overcaffeinated — which feels almost exactly like tired from the inside, restless and urgent and unable to settle, except tired you can fix with a cup, and this you only make worse with one.

Because Wayne has a sweet spot, same as everyone does: enough caffeine to be sharp, not so much that he's rattled. For him it's a band that tops out around 300 mg working at once — under it he's locked in and fast; over it he's exactly where he is now, all engine and no traction. Six cups — the last three crammed into three hours as the crunch hit — sail him clean past it to around 375 mg. And the cruel trick is that the overshoot never feels like "too much." It feels like not enough — so he pours another, and digs the hole deeper.

Caffeolve™ knows the difference he can't feel. As his caffeine climbs toward that line, it doesn't cut him off — it recomposes the cup. The morning cups stay full-strength. But as the tempo picks up and the cups come faster, each one comes back blended a little lighter — enough to hold him at the top of his game and not a milligram past. Six cups, same crunch-driven pace, but the sixth lands him at around 285 mg — sharp, planted in the zone, instead of vibrating past it.

The deadline doesn't move. But Wayne meets it from inside his sweet spot — clear, quick, steady — instead of vibrating through the last stretch getting half as much done. For once the coffee does the exact thing he always asked of it: it keeps him at his best, right when his best is due.

Wayne's ceiling ~300 mg of active caffeine at once. Under it, he's sharp. Over it, wired.

Without Caffeolve™ Six full-strength cups — the last three every hour — push his active caffeine to ~375 mg. Past his ceiling. Jittery, scattered, less productive.

With Caffeolve™ Same six cups, same tempo, blended — active caffeine at ~285 mg. Under his ceiling. Sharp, steady, productive.

Claudia at Three O'Clock

A sensitive user, a late-afternoon slump, and a bedtime ceiling that won't budge. The platform finds the most it can offer.

Most afternoons, around three, Claudia hits a wall. The screen goes a little blurry, the good ideas stop arriving, and there's a pot of coffee down the hall that would fix all of it in about ninety seconds. She doesn't get up. She already knows how this goes: coffee this late stays with her long past bedtime, and she's lain awake on the wrong side of that decision too many times to chance it. So she stays at her desk and grinds through on nothing — a little slower, a little less productive than she'd like to be — and she's been quietly making that same trade every afternoon for years.

It isn't that Claudia stopped loving coffee. It's that somewhere along the line coffee became something she's forced to budget instead of simply enjoying. Her body sets the limit — much past 50 mg still lingering at bedtime and sleep just won't come — and full-strength coffee spends that budget fast. Two cups before mid-morning and she's already near the edge. So the afternoon cup, the after-dinner cup, the second cup with a friend who's only just sat down — those are the ones she gives away, one polite “no thanks” at a time, until her whole coffee life has shrunk to two functional mugs before the day's even started.

Caffeolve™ doesn't touch her limit — her body still owns that 50 mg line. What it changes is the cup — and the confidence to recompose it as the day goes on, with timely guidance at every brew. Caffeine stops being all-or-nothing and turns into something the machine tunes to the hour she's drinking it. Her morning is strong, the way she likes it. And at three o'clock she gets up and pours herself a proper coffee — the warmth, the ritual, the lift to carry her through the afternoon — built so that by bedtime there's nothing left to cost her the night.

Same body. Same line she's never allowed to cross. But for the first time in years, three o'clock isn't a wall Claudia white-knuckles her way over. It's just coffee.

Claudia's ceiling ~50 mg of residual caffeine at bedtime. Under it, she sleeps. Over it, she doesn't.

Without Caffeolve™ Four full-strength cups leave ~**130 mg** at bedtime — more than double her ceiling. To stay under it, she'd have to cut to two cups before mid-morning.

With Caffeolve™ Same four cups, blended — residual caffeine at bedtime ~**49 mg**. Under her ceiling. All four cups, including one at 6 PM.

All values reflect residual caffeine at bedtime (10:30 PM) — each cup decayed from brew time (6-hour half-life).

Diane the Night Before

A board presentation tomorrow. A strong cup today. The dual-ceiling architecture doing exactly what it was designed to do.

Diane has a board presentation at nine tomorrow morning and a slide deck that isn't where it needs to be. It's two o'clock, the afternoon is slipping away, and she heads to the Caffevolve™ device the way most people do at two o'clock on a bad afternoon — looking for something to close the gap between where she is and where she needs to be.

The caffeine recommendation comes back lighter than she was planning on. Not decaf, not even close, but not the full-strength cup she had in mind either. She looks at it for a moment. Then she glances at the numbers the screen is already showing her. The projected bedtime residual tells the story plainly — a full-strength cup right now would still be working through her system well past midnight, and midnight is when she needs to be asleep, not running through tomorrow's talking points in the dark. Her Wake Hours ceiling has room. The Bedtime ceiling doesn't. The platform found the strongest cup that clears both, and it's showing her exactly why.

She smiles. Confirms the brew. The afternoon opens back up, the slides come together, and by the time she shuts the laptop the deck is where it needs to be. She sleeps well. The presentation goes well.

The platform didn't give her less. It gave her the most it could — while keeping tomorrow morning intact.

Two o'clock. Presentation at nine.

Wake Hours ceiling — room to spare. Bedtime ceiling — almost full.

The platform found the strongest cup that clears both.

The Manual Caffeine Adjustment was right there if she'd wanted it. The stronger cup was always her call. She didn't need it.

3. The Adaptive Feedback System

The Caffeine Behavior Instrument is utilized to set the Caffeine Comfort Zone before the first brew. Every answer the user gives during onboarding — how many cups before they feel sharp, how many before they feel wired, the latest they can drink without losing sleep — goes into deriving three personal boundaries. Those boundaries are provisional. They are designed to be close enough to guide the first brew safely, and conservative enough to do no harm while the platform learns. What refines them is the Adaptive Feedback System.

The Adaptive Feedback System (AFS) is the mechanism by which the platform learns from the user's actual experience rather than from what they predicted about themselves at setup. Where the Caffeine Behavior Instrument asks what they expect to feel, the AFS asks what they did feel — after a brew that pushed a boundary, after a night that followed a late cup, after a morning when the user reports they didn't sleep well. Each reported outcome is a signal. Each signal either confirms a boundary or prompts a recalibration.

The Prompt Cadence

The AFS uses a hybrid prompt cadence — not a single trigger mechanism, but three overlapping layers that together ensure every user receives meaningful check-ins, regardless of how closely their brew behavior challenges the Caffeine Comfort Zone boundaries.

The first layer is boundary-triggered: a question fires whenever a brew lands near one of the user's Caffeine Comfort Zone boundaries, at the moment when the response is most informative. A brew that grazes the Wake Hours Caffeine Ceiling is precisely the brew that reveals whether that ceiling is set correctly. A brew timed so that the projected bedtime caffeine residual level approaches the Bedtime Ceiling is the brew that tells the platform most about whether that ceiling needs to move.

The second layer is brew-count-triggered: a proactive floor cadence fires on a rolling basis for each dimension, independent of whether any boundary has been approached. This matters because a pure boundary-triggered system is reactive by design — it only fires when something is near a limit. A well-calibrated user who consistently accepts recommendations and never pushes a ceiling might rarely see a boundary trigger at all. The brew-count cadence closes that gap, ensuring the AFS stays active and continues gathering real-world signal even when the platform appears to be calibrated effectively.

The brew-count cadence has particular importance in the early weeks of use, when the Caffeine Behavior Instrument's provisional boundaries are least refined and the Caffeine Decay Rate is still being established. CDR refinement is the AFS's highest early priority — and the brew-count layer ensures that refinement-relevant questions reach the user on a reliable cadence, not only when a boundary happens to be challenged.

The Three-Layer Prompt Cadence

The AFS uses three overlapping trigger mechanisms to ensure that check-ins reach every user — not just those who push boundaries.

Layer 1 — Boundary-Triggered

A question fires whenever a brew lands near one of the user's Caffeine Comfort Zone boundaries. The user is at the exact moment when their response is most informative — the brew that grazed the Wake Hours Ceiling, the timing that challenged the Bedtime Ceiling.

“Did you feel focused and sharp after that last cup — or a little wired?”

“How did you sleep last night?”

Layer 2 — Brew-Count

A proactive floor cadence fires on a rolling brew-count basis for each dimension, independent of boundary proximity. This ensures that well-calibrated users — those who consistently accept recommendations and rarely graze a ceiling — still receive regular check-ins. The brew-count cadence is also the primary mechanism for CDR refinement in early use, when boundaries are new and the platform needs real-world confirmation regardless of whether any ceiling has been challenged.

Layer 3 — FR-Meta (60-Day)

Every 60 days, a calendar-triggered meta-prompt asks whether the check-in frequency feels right: “Too often,” “Just right,” or “Not sure.” FR-Meta is not tied to any brew event or boundary — it is the long-cycle check that surfaces engagement fatigue and catches slow drift that shorter cadences may miss.

Time-of-Day Anchoring

Both boundary-triggered and brew-count prompts are governed by a second constraint beyond their trigger condition: they only fire at the time of day when the user can answer the question accurately and with minimal friction. Asking a user about the previous night's sleep quality at 3 PM requires them to reconstruct a memory. Asking at the first brew of the morning requires nothing — the answer is already top of mind.

Each of the four feedback dimensions has a designated time-of-day window within which its prompt is eligible to fire. A trigger condition — whether boundary-triggered or brew-count — is only acted upon if the current time falls within the appropriate window for that dimension. Outside that window, the trigger is deferred to the next eligible moment.

The Four Feedback Dimensions

Each dimension captures a distinct aspect of the caffeine experience. Questions are short, one-tap, and timed to the moment in the day when the user's answer is most accurate and most readily available.

D1 — Sleep

"How did you sleep last night?"

→ Refines the Bedtime Caffeine Ceiling. Posed at the first brew of the morning, when sleep quality is immediately top of mind.

D2 — Focus

"Did this brew get you going?"

→ Refines the Peak Zone Floor. Posed mid-morning, after the user has had time to feel the first brew's effect.

D3 — Jitteriness

"Feeling wired or jittery?"

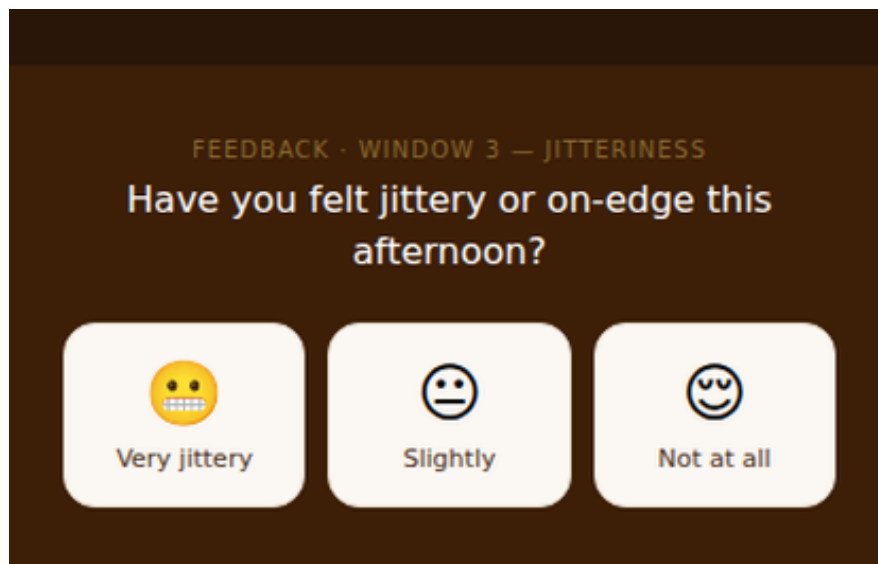
→ Refines the Wake Hours Caffeine Ceiling. Posed mid to late afternoon, when overstimulation effects are most likely to have surfaced.

D4 — Bedtime

"Did caffeine affect how easily you fell asleep?"

→ Refines the Bedtime Ceiling directionally. Posed at the evening brew or last brew of the day, when the user is approaching the window the question is about.

D1 and D4 both inform the Bedtime Ceiling — D1 from aggregate sleep quality, D4 from sleep onset directly. Together they triangulate the boundary from two angles.



AFS — Boundary-Triggered Feedback Prompt (D3 — Jitteriness)

What the Responses Do

Each feedback response is not a direct instruction to move a boundary. It is a signal — one data point that the platform weighs alongside all prior signals for that dimension. The AFS applies an exponential moving average (EMA) methodology: recent responses carry more weight than older ones, so the platform adapts to changes in the user's habits, tolerance, or life circumstances without overreacting to a single outlier response.

In plain terms: the EMA works like a rolling memory that forgets gradually. A response from last week counts more than a response from three months ago, and a cluster of consistent responses in the same direction moves a boundary faster than a single isolated report.

The EMA weighting is also sensitive to boundary proximity: a response collected after a brew that closely grazed a ceiling carries more signal value than a response collected after a brew well within range. Responses gathered near boundaries are therefore weighted more heavily than those gathered far from them, concentrating the refinement signal where it is most diagnostic.

The boundary movement itself is incremental. The AFS moves boundaries in small steps, verifies the effect at the next boundary-grazing brew, and adjusts again if needed. This stepwise convergence avoids the jarring experience of a dramatically different recommendation from one day to the next while still getting the user to a well-calibrated Caffeine Comfort Zone within a few weeks of normal use.

CDR Refinement — The Early Priority

Among the four feedback dimensions, Caffeine Decay Rate refinement is treated as the AFS's highest early priority. CDR refinement begins during Phase 1 Orientation — before any formal prompting starts — as the system passively collects brew timing and residual data from the shared daily log to begin converging the user's decay rate toward their actual biology. The CDR underlies every time-based projection the platform makes — residual level, caffeine headroom, projected bedtime residual — so an inaccurate CDR compounds across every recommendation. The Caffeine Comfort Zone boundaries can be wrong in one direction and the platform still produces useful guidance; a miscalibrated CDR systematically biases every output. The AFS therefore prioritizes decay-relevant signals in the early weeks of use, gathering D1 (Sleep) and D4 (Bedtime) responses on an accelerated cadence so that CDR convergence happens before boundary refinement fully engages.

Boundary Update Rules

Two rules govern every boundary update. First, a minimum response threshold: no boundary moves on fewer than three consistent signals in the same direction. A single response, however clear, is treated as a data point rather than a mandate — the platform waits for a pattern before acting. Second, a maximum step size: no single update moves a boundary by more than a defined increment, regardless of how strong the signal cluster is. This prevents a dramatic shift from one anomalous session and ensures the Caffeine Comfort Zone converges steadily rather than lurching. Together, these constraints mean the platform adapts to the user's actual biology without exposing them to abrupt changes in their daily caffeine experience.

The AFS treats reported outcomes — not observed outcomes — as its input. What the user reports experiencing is the signal, not what the platform’s model predicts they should have experienced. This distinction matters: if the model and the reported experience diverge, the model is questioned, not the user.

AFS enrollment, phase status, and dimension response history are stored at the profile level — not the device level. A household with an Advancara device and five profiles has five completely independent AFS states. One user’s opt-out has no effect on another’s calibration, and one user’s well-refined boundaries have no bearing on the starting point of a new profile added to the same device months later.

The boundary update rules govern how the platform refines the Caffeine Comfort Zone on its own. They do not constrain the user. Any Caffeine Comfort Zone boundary — the Peak Zone Floor, the Wake Hours Ceiling, the Bedtime Ceiling — can be manually adjusted by the user at any time, independent of where the AFS has set it. The platform’s refinements inform the user’s judgment — they never override it.

The Three-Phase Enrollment Arc

The AFS enrolls the user gradually — first observing, then validating and offering, then calibrating — so that participation is earned, not assumed.

Phase 1 — Orientation (approx. Week 1)

The system observes baseline brew behavior silently. No prompts are shown and no user involvement is required. The user is not notified that Orientation is underway. During this period, the system passively collects from the shared daily log: brew count per wake cycle, serving sizes, brew timing patterns, first and last brew times, and any boundary proximity events that occurred naturally during normal use. This data establishes the pre-prompt behavioral baseline — the picture of this user’s real brew behavior before any AFS signal has been collected — against which early feedback signals are calibrated. The Caffeine Decay Rate refinement process also begins during Orientation: every brew contributes timing and residual data that the AFS uses to begin converging the CDR toward the user’s actual biology, even before formal prompting starts.

Phase 2 — Introduction (approx. Start of Week 2)

At the first brew of Week 2, the AFS presents the user with a structured Introduction sequence. This is not simply an enrollment offer — it is a full validation moment comprising four steps presented in sequence:

Step 1 — Observation Summary (P2-OBS). The platform presents the behavioral baseline it derived from Week 1 silent observation: the user’s typical brew count, serving patterns, timing habits, and whether any Caffeine Comfort Zone boundary proximity events occurred. The user sees, for the first time, what the platform has been learning about them.

Step 2 — CBI Validation Questions (P2-VAL). The platform presents a short set of targeted questions mapped directly back to the original Caffeine Behavior Instrument inputs, asking whether the user’s reported onboarding thresholds matched their actual lived experience during the observation week. These include at minimum:

- *“Over the past week, did the number of servings it took before you felt alert and focused match what you reported when you set up your profile?”*
- *“Did your overstimulation threshold — the point where you felt wired or jittery — feel consistent with what you told us?”*
- *“Was your reported caffeine cutoff time accurate, or did you find yourself brewing later without it affecting your sleep?”*

The answers to these questions are used immediately to flag any CBI-derived boundary that the user’s lived experience suggests may be miscalibrated — marking those boundaries as higher-priority refinement targets for Phase 3.

Step 3 — Boundary Review and Adjustment (P2-BDY). The platform presents the user’s current Caffeine Comfort Zone boundaries — Peak Zone Floor, Wake Hours Caffeine Ceiling, and Bedtime Caffeine Ceiling — and invites the user to review and adjust any value before AFS prompting begins. This is the user’s opportunity to correct any provisional prior that their first week of experience has revealed as inaccurate. Adjustments made here are applied immediately. Phase 3 Calibration then begins from this user-validated baseline rather than the raw CBI provisional prior.

Step 4 — Enrollment Offer (P2-ENR). With the observation summary reviewed, validation questions answered, and boundaries confirmed or corrected, the platform presents the AFS enrollment offer: a brief explanation of what check-in prompts involve, and two options — “Count me in” or “Not right now.” The user opts in with complete visibility into what the platform has observed and with boundaries they have confirmed or corrected. If the user declines, the three-stage opt-out arc begins.

Phase 3 — Calibration (ongoing)

Once enrolled, the user receives boundary-triggered and brew-count-triggered prompts for each feedback dimension according to the cadences described above. Refinement proceeds from the user-validated baseline established in Phase 2 — not from the raw CBI provisional prior. Questions fire only when a brew lands near a relevant Caffeine Comfort Zone boundary, or on the brew-count proactive floor cadence. As responses accumulate, the provisional boundaries give way to boundaries shaped by how this specific person actually experiences caffeine.

The Opt-Out Arc

AFS participation is always voluntary. The three-stage opt-out arc is designed to distinguish temporary disengagement from genuine objection. The first opt-out pauses prompts and presents a re-enrollment offer — the user may simply have found the timing inconvenient. The second opt-out pauses prompts for a longer window. The third opt-out is permanent: prompts are disabled for the profile and do not resume. The device operates normally under the stored Caffeine Comfort Zone boundaries regardless of AFS enrollment status. The platform does not require ongoing feedback to function — it benefits from it.

Opt-out is reversible at the first and second stages — the user can re-enroll at any time, and all accumulated signal history is preserved when they do. No prior data is discarded on opt-out; only the prompting is paused. A user who opts out after six weeks of calibration and re-enrolls a month later returns to their accumulated history intact, with refinement resuming from where it left off rather than starting over.

The Science Advises. The User Decides.

Where reported outcomes diverge significantly from what the platform's model predicts — a gap the system recognizes as a 50% variance trigger — the AFS does not quietly recalibrate. It surfaces the discrepancy as an informed-choice advisory. The user is told, plainly, that the platform's working model and their reported experience don't match, and is invited to reflect. This is the AFS expression of the platform's core Advisory Philosophy: the science advises. The user decides.

The platform does not override the user's lived experience with a model. It flags the discrepancy honestly and leaves the next step to the user.

When Answers Need More Context

Some feedback responses touch more than one dimension. A user who reports poor sleep and high jitteriness on the same day is providing correlated data that touches both the Bedtime Ceiling and the Wake Hours Ceiling. The AFS is designed to recognize these co-occurrences and to weight them accordingly rather than routing each answer in isolation to a single dimension. In cases where a response is ambiguous — for example, poor sleep that could be explained by caffeine or by any number of other factors — the AFS assigns lower confidence to the update and requires more confirming responses before moving the boundary.

When the Caffeine Behavior Instrument's initial answers were themselves contradictory or at the edge of what the back-calculation can resolve — for example, a user who reports needing very few cups to feel alert but also very few cups before feeling wired — the AFS enters a higher-sensitivity early-calibration mode. Boundaries that were set with lower confidence at onboarding are marked as higher-priority refinement targets, and the AFS prioritizes questions in those dimensions first.

The Role of Scientific Research

The Caffevolve™ platform is grounded in peer-reviewed pharmacokinetic research on caffeine metabolism and its effects on sleep architecture. That science informs the decay model, the bedtime ceiling values, and the Reality Range variance band. But in the AFS — as in the rest of the platform — science is consultative, not constraining. The research tells the platform what to expect. The user's reported experience tells it what is actually happening for this specific person.

When a user's reported outcomes diverge from what the pharmacokinetic model would predict — when a 50% variance from model expectations is triggered — the AFS surfaces this as an informed-choice advisory rather than silently overriding either the model or the user. The user is given the information they need to make a considered decision: the platform's model, what their reported experience suggests, and the option to explore whether a different boundary setting would serve them better. This is not a fault condition. It is the system working as designed — honest about uncertainty, respectful of individual variation, and always deferring the final call to the person holding the mug.

FR-Meta: The 60-Day Check-In

Every 60 days, FR-Meta — the third layer of the AFS prompt cadence — presents a single meta-prompt asking whether the check-in frequency feels right. Three options: “Too often,” “Just right,” “Not sure.” The purpose is simple: surface engagement fatigue before it leads to opt-out. FR-Meta fires on a calendar cadence — not tied to any brew event or boundary trigger.

4. The User Profile

A significant amount of information about the user is needed for platform intelligence to build a caffeine recommendation each time a brew is initiated. The responses the user provides to the Caffeine Behavior Instrument (CBI) are a large contributor to this information base. If the user was required to input all of this information every time they wished to brew a cup, the process would likely be quite time consuming and tedious. Given that virtually all of this information is relatively stable — and not changing often — it can be trusted for relatively long periods of time, and recalled with a single tap each time the user commences a brew process. This is a much more positive user experience and eliminates the possibility of inaccurate information being entered. This is the purpose of the User Profile.

When a user decides to initiate a brew, they simply select their Profile, enter an expected bedtime for that day, and a caffeine recommendation can be generated. It requires about five minutes to initially set up the User Profile, but once completed, requires only about five seconds of the user's time to get a brew started. Efficient and convenient.

Included in the Resident User Profile are the parameters of the user's personal Caffeine Comfort Zone — their Peak Zone Floor, Wake Hours Ceiling, and Bedtime Ceiling — along with their personal Caffeine Decay Rate. In addition, a preferred grind setting and list of preferred additives are saved.

Optional additions to the Resident User Profile include setting up Sub-Profiles and a BrewNow! Profile. A user can build and save up to 3 individual Sub-Profiles on Originara and Culminara Office, or up to 6 on Advancara — routines that are tied to a particular day and time window, such as a weekday-morning boost or a weekend wind-down blend. When a brew is initiated while a Sub-Profile window is open, the platform automatically applies the chosen brew strength and additives identified in the active Sub-Profile. A BrewNow! Profile saves a particular serving size, serving count, brew strength setting, and additive(s) that are automatically applied when the express BrewNow! brew path is chosen.

Resident and Guest Profiles

There are two types of Profile recognized by the Caffevolve™ Platform: a Resident User Profile and a Guest Profile. The preceding description is that of a Resident User Profile — a resident of the household or office, or a frequent user of the device, who utilizes the Caffevolve™ brewing device on a regular basis. Given that the number of user profiles that can be created is limited, it is prudent to restrict these profiles to frequent users of the brewing device. The number of Resident User Profiles supported scales by tier: Originara supports 2, Advancara 5, and Culminara Office 20.

A Guest Profile is automatically established when a guest completes the Quick Profile Path — name, initials, a self-selected caffeine sensitivity level, and a declared bedtime. The Guest simply selects a caffeine sensitivity level based on their own gut feeling rather than having it determined through their responses to the Caffeine Behavior

Instrument (CBI); the Guest Profile is established with known average parameters to establish a personal Caffeine Comfort Zone, while the Resident User Profile benefits from the personalized parameters set by completing the CBI process. At the end of the Quick Profile Path, the guest is asked whether they are likely to brew again: if so, the profile is saved as a Returning Guest Profile, loading and saving settings from the guest's last completed brew (size, strength, additives) for reuse via the Returning Guest Path; if not, the profile is used for that single brew only and then discarded.

Guest Sensitivity Presets — Guest profiles only

Selected on the Quick Profile Path. Not applied to Resident Users.

SENSITIVITY	WAKE HOURS CEILING	BEDTIME CEILING	PEAK ZONE FLOOR	CAFFEINE DECAY RATE
Higher	200 mg	30 mg	40 mg	7.0 hr
Moderate	300 mg	50 mg	40 mg	6.0 hr
Lower	400 mg	70 mg	60 mg	5.0 hr

These fixed presets are a deliberate conservatism shortcut for guests, who do not complete the Caffeine Behavior Instrument. Higher sensitivity pairs the lowest ceilings with the slowest assumed clearance (7.0 hr) — both the more protective setting — so the two move in the same direction. Resident Users receive no tier: their boundaries are computed individually by the CBI from reported behavior.

5. The Device

Section 5 covers the CaffevolveTM device hardware — the components that physically execute the platform’s calculations. All of it is made possible by a single foundational architectural decision: blend-then-grind. Before any hardware component is described, that decision deserves its own introduction — because everything the device does flows from it. The Hardware subsection then covers the components that execute the blend-then-grind architecture: the dual-hopper system, gravimetric dispensing, blending chamber, grind system, water reservoir, carafe, and firmware. The touchscreen interface through which the consumer interacts with the device — the profile-driven brew path a registered user follows from the Brew Gateway to a completed cup, and the guest path for users without a stored profile — is covered separately in Section 7, The User Experience.

The Blend-Then-Grind Architecture

The foundational CaffevolveTM invention is blend-then-grind: caffeinated and decaffeinated beans are blended in the system-computed ratio before they reach the grinder, not after. This single architectural decision is what makes precise caffeine management from a single brewing device possible at all.

Pre-grinding blending is the conventional alternative — mixing pre-ground caffeinated and decaffeinated coffees in some proportion. It is also an unreliable one. Caffeine distribution in a pre-ground blend depends on how uniformly the discrete caffeinated and decaffeinated particles mix before and during brewing, a variable that cannot be controlled with precision. Blend-then-grind eliminates that uncertainty at the source.

Why Blend-Then-Grind Changes Everything

The blend-then-grind method establishes the caffeinated-to-decaffeinated ratio at the whole-bean level, before the grinder sees the beans. Every particle of ground coffee that results contains both bean types in the target proportion — the blend is homogeneous at the particle level. This matters because caffeine extraction is a particle-level event. If the blend is established after grinding — by mixing pre-ground caffeinated and decaffeinated coffees — the result is a physical mixture of discrete fully-caffeinated and fully-decaffeinated particles. Caffeine distribution then depends on how well those particles mix before and during brewing, which is inherently variable and unpredictable. Blend-then-grind eliminates that variable entirely. Because every gram of ground coffee carries the same caffeine load, the Blend Ratio Formula’s output is reliably accurate: the caffeine yield is consistent regardless of particle settling, channeling, or brewing variation. The math and the cup agree.

This is the foundational CaffevolveTM invention — and the reason precise caffeine management from a single device is possible at all.

5.1 The Hardware

5.1a. Brew Components

The Dual-Hopper System

The device stores caffeinated and decaffeinated beans in two separate reservoirs: Hopper A, which holds the caffeinated beans, and Hopper B, which holds the decaffeinated beans. The dual-hopper architecture enables real-time blend ratio adjustment without any manual bean mixing — the system independently controls how much it draws from each hopper on every brew.

During device setup, the user declares the Bean Family Classification (Arabica or Robusta) and Roast Level Classification of the beans loaded in each hopper. Bean Family Classification determines the bean family caffeine multiplier applied to all subsequent caffeine calculations: Arabica is the baseline; Robusta carries approximately 1.8 times the caffeine content. Roast Level Classification is advisory for pairing consistency and has no impact on caffeine calculations. These declarations are stored as the Hopper Content Configuration and require Admin PIN access to modify after initial setup.

Twin Bean Certified Pairings

A Twin Bean pairing is a caffeinated and decaffeinated bean matched and certified together through the Caffevolve™ Twin Bean Program — a planned Caffevolve™ initiative developed in partnership with premium coffee roasters. Rather than treating the two beans as independent, the Program selects and characterizes them as a set: chosen wherever possible from a shared origin — the same growing region, ideally the same farm — so the caffeinated and decaffeinated components begin from a common flavor foundation, then each roasted to its own calibrated profile to account for the different way the two behave under heat. The result is a pair engineered to taste as though it belongs together at every blend ratio, so that a fully caffeinated morning cup and a decaf-leaning evening cup read as two expressions of the same coffee rather than two different ones.

Certification also captures the pairing's measured characteristics as a set: bean family, roast level, and caffeine density for both the caffeinated and decaffeinated component. Those values are what the blend calculation consumes, and a certified pairing supplies them directly rather than relying on the user's declaration at hopper load. A Twin Bean Certified pairing is therefore the configuration in which cup consistency across the blend range is fully optimized, and in which hopper setup requires no manual entry at all.

Gravimetric Dispensing

The device dispenses beans by weight, not by time or volume. A precision scale measures the mass dispensed from each hopper in real time, stopping the auger the moment the target weight is reached. This matters because bean density varies — by roast level, by fill level, by ambient humidity — and a time-based or volume-based dispenser would introduce error that compounds across every brew.

Gravimetric dispensing is the mechanical link between the Blend Ratio Formula and the physical cup. The formula computes exact gram weights for the caffeinated and decaffeinated fractions; gravimetric dispensing delivers them. Without it, the mathematical precision of the blend calculation would not survive contact with the real world.

The Blending Chamber and Agitator Arm

After the hoppers dispense their respective bean loads, the beans drop into the blending chamber — the internal cavity where the blend ratio is physically realized before grinding. Passive mixing alone is insufficient: bean density and size differences between caffeinated and decaffeinated varieties mean gravity would produce an uneven distribution. The blending chamber addresses this with an active mechanical solution.

A stepper motor drives a reciprocating agitator arm through a controlled 180° arc, completing approximately five to six sweep cycles over two to three seconds. The result is a uniformly intermixed bean load in which caffeinated and decaffeinated beans are distributed evenly throughout the chamber before the combined mass drops into the grinder. The agitator arm design is standardized across all product tiers.

The Grind System

Ground particle size governs two distinct extraction measures at once: the solids yield that determines flavour (the SCA 20–22% band) and the caffeine extraction efficiency that the Blend Ratio Formula uses as a direct input — the proportion of the caffeine contained in the dosed beans that reaches the cup (0.6277 at the medium-grind reference). The two move together with grind but are denominated differently and are not interchangeable. Because the formula depends on caffeine extraction efficiency specifically, grind consistency is not just a quality concern; it is a calculation-accuracy concern. The device controls grind through a calibrated burr mechanism whose settings map to known values for both measures.

Grind setting is stored at the profile level and applied automatically at brew time. The user selects their preferred grind during profile setup; the device references the appropriate caffeine extraction efficiency at every subsequent brew without requiring the user to manage the relationship between grind and caffeine output.

The grind setting alone determines caffeine extraction efficiency; the blend ratio does not affect it. Because blend-then-grind establishes the caffeinated-to-decaffeinated ratio at the whole-bean level, both bean types enter the burrs together and fracture identically under the same grind setting. The resulting grounds are homogeneous at the particle level, so the particle-size distribution — and with it the extraction efficiency the Blend Ratio Formula assumes — is the same at any blend ratio. A single stored grind setting holds across the full range of Caffeine Blend Recommendations the day may call for, with no per-brew adjustment required.

The Water Reservoir

Every brew begins with water, and how much water the device holds — and how fresh it is — is a deliberate part of the design rather than an afterthought. On each consumer tier the reservoir is sized to hold an estimated full day

of brewing on a single fill, and it is removable: the user carries it to the sink to refill and returns it to the device. Two goals shape it at once. It must hold enough for the day's estimated brew load, and it must stay light enough to carry comfortably when full.

Sizing the Reservoir

Reservoir capacity is built up from a single unit: the 12 oz serving. Producing a 12 oz cup draws more than 12 oz from the reservoir, because the spent coffee bed retains water it never releases and a small amount is lost to evaporation over the brew. Accounting for both, the device draws approximately 14 oz of water for every 12 oz served. Capacity from there is simply the number of servings a tier is estimated to deliver in a day, multiplied by that per-serving draw.

Reservoir Capacity by Tier

- **Originara** — 56 oz (approximately 4 servings), roughly 4.3 lb when full
- **Advancara** — 84 oz (approximately 6 servings), roughly 6.0 lb when full
- **Culminara (COMMERCIAL)** — 90 oz batch capacity, sized for continuous throughput rather than a single daily fill

Filled weight is treated as a design constraint, not just an outcome. Originara's full reservoir carries at roughly 4.3 lb and Advancara's at roughly 6.0 lb — both kept within comfortable one-handed carry — and each is supported by a positioned handle and a wide fill opening so the tank is easy to grip and quick to fill.

Culminara is a commercial product, built for cafes, offices, and hospitality venues. Its reservoir is a throughput batch replenished continuously across a service day rather than filled once each morning, so the daily-fill sizing logic of the consumer tiers does not apply to it, and constant turnover keeps its water fresh as a matter of course.

Why Fresh Water

Water that sits for days in a reservoir goes stale. It picks up off-flavors from the air and from the tank itself, and standing water is simply not the starting point a good cup deserves. The remedy is a reservoir designed to be refilled at the start of each day, and it also reminds the user when water has been standing too long. Fresh in, fresh out.

Two Reservoir Reminders

The reservoir carries two independent reminders. They answer different questions and behave differently — one gates a brew, the other only advises.

Water Sufficiency — a hard gate. Before every brew, the device compares the water in the reservoir against what the requested brew requires, reading level from the reservoir's water-level sensor. If the reservoir holds enough, the

brew proceeds. If it does not, the device does not begin — it alerts the user that water must be added first. This is the platform's one hard water gate: a brew that cannot be completed with the water on hand does not start, so the user is never left with a half-filled carafe.

Water Age — a soft advisory. Independently, the device tracks how long the current water has been standing. The moment a reservoir is refilled, an age clock starts; if that water is still in the tank 36 hours later, the device surfaces a freshness reminder — a suggestion to empty and refill, never a block. The user can always brew with the water they have. The platform advises; the user decides.

The two reminders do not interact, and that independence is deliberate. Water level never suppresses the age reminder. A reservoir can be completely full and still be flagged as old — which is exactly the case that matters most. A user who has been away for several days returns to a full tank of water that has been standing the entire time; because only a refill resets the age clock, not the water level, that full, stale tank is correctly flagged for a refresh. A full reservoir is not evidence of fresh water. Only a recent refill is.

The First-Brew Freshness Reminder

At the first brew of the user's day — whenever that falls, since the platform is schedule-agnostic and a user's day may begin at any hour — the device offers a brief reminder that fresh water makes a better cup. When that first brew also lands past the 36-hour mark, the two messages combine into a single, stronger prompt rather than two separate nudges.

What Counts as a Refill

An age-clock reset requires a genuine refresh, not a splash. Topping a small amount of water onto a tank of older water does not restart the clock — if it did, a few ounces added to two-day-old water would misrepresent the whole reservoir as fresh. Using the level sensor already present, the clock resets only when a fill brings the reservoir up to near-full. A true refill resets the clock; a partial top-up does not.

Filling to Any Level

The reservoir is sized for a full day, but the user is never required to fill it to the top. Any fill level below capacity brews normally, because the pre-brew sufficiency check confirms adequate water before every brew — a user who prefers a lighter reservoir can fill partway and rely on the device to catch any shortfall before a brew begins. Graduated capacity markings on the tank let the user fill to a known serving level on purpose.

The Carafe

The carafe is the final stage of the brew path and the vessel in which in-stream additives — dispensed after extraction — complete their dissolution. It is where the day's coffee is held, and its design reflects two priorities: keeping that coffee hot without degrading it, and making the additive mixing that happens inside it easy to keep clean.

An Insulated Carafe on Every Tier

Every Caffevolve™ carafe is vacuum-insulated — double-walled, with no glass carafe and no warming plate on any tier. The reason is quality. A glass carafe holds temperature by sitting on a heated plate, which slowly scorches the coffee and flattens its flavor the longer it sits. Vacuum insulation holds temperature passively instead, keeping the coffee hot for hours without a burner and without cooking it. This also draws no standby power, and it reinforces what the platform is built around: a cup brewed earlier is still a good cup later, so an afternoon pot is still worth pouring in the evening.

Carafe capacity is matched to each tier and enforced by the device at brew time, which prevents a serving-count selection that would exceed the installed carafe:

Carafe Capacity by Tier

- **Originara** — approximately 40 oz insulated
- **Advancara** — approximately 60 oz insulated
- **Culminara (COMMERCIAL)** — 90 oz insulated server

The Removable Baffle System

As brewed coffee flows into the carafe carrying its dispensed additive, the carafe's internal baffle geometry sets the coffee spinning — a turbulent, helical path that mixes the additive through the full volume and drives its dissolution. Every tier carries this passive baffle; it is the mechanism that turns a dose dropped into the stream into an evenly integrated cup.

What makes the Caffevolve™ baffle different is that it lifts out. A baffle fixed to the carafe wall traps exactly what this device dispenses — coffee oils, and the proteins, collagen, and oil-based wellness additives that dry into films a brush can't reach in a fixed corner. So the baffle is built as an independent part: a central column with helical vanes that seats in the carafe, structurally separate from the carafe wall, and lifts out as a single piece for cleaning. The vane edges stand off from the wall rather than meeting it, so nothing is trapped between the two — lift the column out and the carafe interior is a smooth, open vessel that wipes clean in one pass, while the column itself rinses clear under the tap.

The column is designed to be effortless in daily use. It locks into a socket in the carafe floor with a quarter-turn — twist to seat it before brewing, twist to release it for cleaning — so the incoming brew stream can't unseat it. It carries a finger loop at the top for lifting. And because the helical column is symmetric about its axis, there is no wrong way to put it back: dropped in at any rotation, the mixing geometry is the same. Coffee pours around the seated column to the spout, so the user removes it only to clean, never to serve.

Flex-Vane Vanes for the Insulated Neck

An insulated carafe has a narrower neck than a glass one — the double wall requires it — and a mixing column wide enough to work well in the body would be too wide to pass back out through that neck if its vanes were rigid. The Caffevolve™ column solves this with flexible vanes on a rigid central spine. As the column is drawn up through the neck, the neck presses the vanes inward and they bow toward the spine, narrowing enough to clear the opening; once past the neck, they spring back to full span. Reinsertion reverses it. The neck does the work — there is no lever, collar, or fold step for the user to operate, and because the vanes flex rather than pivot, the column has no hinges or joints of its own to trap residue. One baffle design serves every tier this way.

Culminara: Ultrasonic Assist

Culminara adds an active 40 kHz ultrasonic pulse to the same removable baffle geometry. The higher additive volumes and continuous throughput of a commercial setting benefit from the extra dissolution energy; the ultrasonic system lives in the carafe base, beneath the seated column, and works alongside the passive baffle rather than replacing it.

Care

The carafe and its baffle column are kept clean with a warm-water rinse after daily use, with a periodic warm-water-and-vinegar soak — column expanded and in place — to clear accumulated oils and films. Because the column lifts out and its vanes carry no joints, both the carafe interior and the column present only smooth, open surfaces to that routine.

Firmware

The device's onboard firmware is the operational layer that ties the physical hardware to the intelligence platform. At every brew event, firmware runs the full eight-step dual-ceiling calculation, retrieves the current residual caffeine level, applies the Blend Ratio Formula, and translates the result into precise dispense instructions for the gravimetric system. It manages the brew cycle sequence — hopper dispensing, blending chamber agitation, grinding, brewing, and synchronized additive dispensing — and maintains the brew log that the companion app reads and writes against.

Firmware updates are delivered over-the-air and are managed at the device level. The calculation engine, the additive library, and the platform's pharmacokinetic constants are all firmware-resident, ensuring that every device in the field operates from a consistent and current version of the platform logic.

Every calculation the platform makes is only as good as the machine that executes it. The blend recommendation produced by the calculation engine specifies a precise caffeinated-to-decaffeinated ratio — and the device's job is to deliver that ratio in the cup, brew after brew, with the accuracy the math demands.

5.1b. The Additive System

The hopper system manages caffeine. The additive system manages everything else the user chooses to put in their cup. Together they make the Caffevolve™ device more than a precision caffeine brewer — they make it a configurable daily supplement and flavor platform, all operating from a single brew cycle.

Canisters and Slot Codes

Where the device uses hoppers for beans, it uses canisters for additives. The two terms are not interchangeable: hoppers are the large dual bean reservoirs; canisters are the smaller refillable containers that hold powdered additives. Every device accommodates two canister families, each with its own dedicated slot code prefix: Flavor canisters occupy F-coded slots, and Wellness canisters occupy W-coded slots.

The number of canister slots scales with device tier, while the two-family Flavor / Wellness organization is retained across every tier.

Canister Configuration by Tier

- **Originara** — 3 canisters (2 Flavor: F-1, F-2; 1 Wellness: W-1)
- **Advancara** — 6 canisters (3 Flavor: F-1–F-3; 3 Wellness: W-1–W-3)
- **Culminara Office** — 12 canisters (6 Flavor: F-1–F-6; 6 Wellness: W-1–W-6)
- **Culminara Caffè** — a greater number of canisters for broader self-serve service; count being finalized

Each canister carries a permanent alphanumeric slot code imprinted at manufacture — for example, F-1 for the first Flavor slot or W-2 for the second Wellness slot. The slot code is the canister's identity within the device and is used to register, track, and display its contents. Slot codes do not change when a canister is refilled or re-registered; the code identifies the physical slot, not the ingredient currently loaded in it.

Canister family segregation is enforced through three reinforcing layers. First, physical keying: F-family and W-family canister slots are physically shaped to accept only their corresponding canister family — a W canister cannot be inserted into an F slot by mechanical design alone, requiring no sensor or firmware intervention. Second, color coding: F-family canisters and W-family canisters are manufactured in distinct colors, providing immediate visual differentiation before insertion is attempted. Third, alphanumeric slot codes: each canister bears a permanent human-readable slot code comprising a category prefix (F for Flavor, W for Wellness) and a sequential slot number, imprinted at manufacture, providing precise identification during registration and on the device's canister shelf display. Together these three layers ensure that cross-contamination between Flavor and Wellness additive families is prevented at the physical design level, independently of any software or firmware control.

The Two-Auger Dispensing System

The device uses a two-auger dispensing system: one auger dedicated to the Flavor canister family, one to the Wellness canister family. The separation is intentional — keeping Flavor and Wellness on independent auger pathways prevents cross-contamination between additive categories, regardless of which specific canisters are installed or how frequently they are swapped.

Within each family, a microcontroller-driven canister selection mechanism routes the appropriate canister's output to the shared auger pathway. Selection is entirely automatic — the user chooses their additive on the touchscreen, and the device handles the mechanical routing without any manual valve operation.

Dispensing is synchronized with the active brew flow. After extraction, as hot brewed coffee moves from the brew chamber toward the carafe, the auger meters a calibrated dose from the selected canister and delivers it to a dispensing port positioned in the coffee flow path. The hot, actively moving coffee entrains the powder and carries it into the carafe, where the baffled carafe geometry promotes turbulent mixing and dissolution. Dispensing into an active flow — rather than into a finished, static beverage — maximizes both dissolution efficiency and ingredient integration.

The Additive Library

The device does not permit unrestricted additive loading. Every additive that can be registered to a canister must appear in the Additive Library — a curated, Caffevolve™-validated on-device database of approved Flavor and Wellness ingredients. Library entries have been tested for compatibility with the dispensing system across four physical characteristics: particle size, density, flow rate, and moisture behavior. An ingredient that cannot be reliably metered, dispensed, and dissolved under in-stream conditions is not listed.

The Additive Library is filtered by canister family at the point of registration: when a user registers an F-slot canister, they see only Flavor entries; when registering a W-slot canister, only Wellness entries appear. The library is a Caffevolve™-managed asset, updatable by Caffevolve™ or its licensees as new validated ingredients are approved. Custom or user-added additives outside the library are not permitted.

Canister Registration

Registration is triggered automatically on canister insertion. Each canister slot incorporates a slot-level insertion sensor that detects when a canister has been successfully seated. Because the physical keying ensures that only a correct-family canister can be seated in a given slot, successful insertion confirms correct-family placement without any electronic verification. The device identifies which slot was triggered by the sensor and immediately presents a category-filtered registration prompt on the touchscreen — showing only additive library entries in the family corresponding to that slot. The user selects the specific ingredient loaded from the filtered library view. Once confirmed, the registration is stored and displayed on the canister's digital label.

The slot code is a human-readable label, not a machine-readable identifier. The device does not read or scan the slot code — it detects insertion via the slot sensor and relies on the physical keying to confirm family correctness.

The slot code's role is to identify the canister to the user and to the device's display system, not to communicate with the device electronically.

Registration does not expire on its own. If the canister is refilled with the same ingredient, no re-registration is needed — only a refill confirmation to reset the fill-level counter. If the ingredient changes, the user initiates a new registration, which replaces the prior record for that slot.

Compound Categories and the Dissolution Hold Timer

Different additive compounds dissolve at different rates in hot coffee. A flavoring powder engineered for rapid dissolution in a hot beverage application behaves very differently from a protein supplement or an oil-based extract under the same conditions. If the device dispensed every additive the same way and moved on immediately, undissolved powder would reach the carafe — affecting both the beverage and the user's experience of the additive's intended effect.

The platform addresses this through the additive compound category system. Each Additive Library entry is assigned to one of six compound categories based on its chemical composition and physical dissolution characteristics. Each category carries a scientifically derived minimum hold time — the duration required for that compound type to fully dissolve in hot coffee under active flow conditions. When an additive is dispensed, the Dissolution Hold Timer reads the compound category of the selected ingredient and applies the appropriate hold duration automatically. The user never manages hold times directly.

The Six Additive Compound Categories

Each entry in the Additive Library is assigned to one of six compound categories. The category determines the dissolution hold time applied automatically after dispensing.

Proteins & Peptides — Longest hold time. Dense molecular structures require extended exposure to hot liquid for full dissolution.

Fats & Oils — Extended hold time. Emulsification into hot coffee is time-dependent; turbulence from baffled carafe assists.

Plant Extracts — Moderate hold time. Dissolution varies by extract density and particle fineness.

Vitamins & Minerals — Moderate hold time. Water-soluble vitamins dissolve readily; mineral compounds may require slightly longer.

Amino Acids — Short hold time. Highly water-soluble; dissolve rapidly in hot liquid under active flow.

Flavor Additives — Shortest hold time. Fine-particle powders engineered for rapid dissolution in hot beverage applications.

Hold times are stored as properties of each Additive Library entry and applied automatically — the user selects the additive, the system determines the appropriate hold duration.

Fill-Level Monitoring

The device tracks the remaining content of each registered canister through a user-declared initialization and firmware dispense-event decrement system. Each additive canister incorporates six molded fill lines on the canister body, labeled 1 through 6 from bottom to top, providing a user-readable reference scale. At registration or refill confirmation, the user selects the fill line nearest to the powder level in the canister — declaring the starting fill level for that slot. No weighing mechanism is required.

From the declared starting level, the device tracks remaining content by decrementing the fill-level counter by the known dose weight of the registered ingredient on each dispense event. The three-state indicator — Green, Amber, Red — is computed independently from this running estimate as a percentage of full canister capacity: Green above 25%, Amber between 10% and 25%, Red below 10%. The fill-level scale and the indicator thresholds are independent — a user who declares level 3 of 6 starts in Green and crosses into Amber and Red at calculated thresholds, as does a user who declares level 6 of 6, just later.

The fill-level counter resets upon re-registration following a refill of the same ingredient, or initializes at the user-declared starting level when a new ingredient is registered. This ties fill-level tracking directly to the registration events established by the slot-triggered registration system.

Fill status is surfaced at two points: on the canister's digital label during normal operation, and as a pre-brew refill alert on the brew configuration screen when an Amber or Red additive is selected for a brew. The alert appears before the brew is confirmed, giving the user the opportunity to refill before the session begins.

Canister Fill-Level Indicators

Green — >25% estimated remaining **Amber** — 10–25% estimated remaining **Red** — <10% estimated remaining

Note: these thresholds apply to the firmware's running estimate based on user-declared starting level and cumulative dispense-event decrement — not a direct weight measurement.

Filling Funnels and Refill

Canisters are refillable, and refilling is designed to be clean and unambiguous. Two dedicated filling funnels are included with every device — one for the Flavor family, one for the Wellness family — each color-matched to its canister family. Both are stored onboard the device in a dedicated compartment adjacent to the canister shelf, so the correct funnel is always at hand at the point of use.

Two Filling Funnels

- **Flavor funnel** — warm tone, color-matched to Flavor canisters; seats into the open sliding top of a Flavor canister.
- **Wellness funnel** — cool tone, color-matched to Wellness canisters; seats into the open sliding top of a Wellness canister.
- **Onboard storage** — both funnels are held in a dedicated compartment adjacent to the canister shelf.

Each canister carries a sliding top that seals its contents during transport and slides open to expose the fill opening. Canisters are removable and are filled on the counter rather than in place: the user pulls the canister from its slot with the sliding top closed, retrieves the matching funnel, sets the canister on a flat surface, slides the top open, and seats the funnel — which seats positively into the open top — before pouring. The color-matched funnel and the family-specific keying together keep Flavor and Wellness additives from crossing during a refill.

When the canister is returned to its slot, the slot's insertion sensor detects the reinsertion and the device prompts the user to confirm the existing contents or register a change — the same registration flow triggered on first insertion.

Additives and the Sub-Profile System

Additive selections are a dimension of personalization entirely separate from caffeine management. They are stored at the sub-profile level — each sub-profile holds its own additive canister selections and dose levels, independent of the blend ratio, which is always computed fresh at brew time from the caffeine calculation. A user can configure one sub-profile for a morning wellness routine with a specific Wellness additive and a separate sub-profile for an afternoon brew with a different Flavor additive, without either configuration affecting the other or the underlying caffeine recommendation.

When a sub-profile window is active at brew time, its additive selections become the default for that brew — including when the user initiates via BrewNow!. Additive selections can always be adjusted at brew time regardless of sub-profile defaults. The additive system is designed to be present when the user wants it and invisible when they don't.

6. CaffevolveGo

CaffevolveGo is the platform component that extends the Intelligence Center's reach beyond the device, into wherever the day actually takes the user. Every cup consumed outside the device is a cup platform intelligence would otherwise never know about; CaffevolveGo closes that gap, through two components: External Caffeine Entry (ECE) and the Offsite Advisory System (OAS).

Both are reached through a single entry point. Opening CaffevolveGo presents the two functions as one branded choice — *External Caffeine Entry* to record a drink already consumed, or the *Offsite Advisory System* to weigh what to consume next — so the user selects the task directly rather than navigating to a separate screen for each. The choice becomes available once a bedtime is set for the session, since the OAS projection depends on it.

External Caffeine Entry (ECE)

ECE lets the user quickly log every external drink consumed, keeping the platform's model of their caffeine intake accurate and up to the moment. Every external caffeine source is logged and fed back into the Caffeine Comfort Zone model, keeping the platform's intelligence complete across the full day. This ensures that when the user looks to brew or purchase another cup later in the day, the caffeine recommendation is always based on a complete picture of all they have consumed since waking. The device manages the cup at home; CaffevolveGo, through ECE, manages it everywhere else — closing the 24-hour window by capturing everything platform intelligence needs to know about the user's day, wherever it happens.

The Offsite Advisory System (OAS)

OAS is the second component of CaffevolveGo, and the one that puts the user's current caffeine standing directly in their hand when they are away from home — displayed in real time, so every decision away from home is as informed as every decision at it. Rather than simply logging what has already been consumed, OAS presents actual drink options against the user's current standing, turning the record of what's already been consumed into guidance for what to consume next.

OAS organizes these options by venue, since what is available — and how much control the user has over it — changes with the setting. The workplace is the venue users encounter most consistently, and OAS treats it as the primary setting: *Worksite*. The drinks there tend to be a shared office pot, single-serve pods, tea, and packaged or canned drinks, and the strength of a communal brew is fixed rather than chosen. OAS accounts for this by making serving volume the lever at the workplace — a full, two-thirds, or one-third pour — and by letting a user save the specific drinks they return to as personal workplace options, so that on each visit those recurring choices are presented and measured against their current standing.

Away from the workplace, OAS draws on its own database of venue-specific beverages for cafés and convenience stores — presenting the estimated caffeine content of all likely available drinks at that venue alongside the user's

real-time caffeine and headroom status. The screen-level behavior of all three venues is detailed in Section 7.

Both components share the same Intelligence Center that drives the brewing device. Whatever the user chooses is sent immediately back to the Intelligence Center, logged and folded into that day's caffeine calculations, so the next recommendation at home remains accurate.

7. The User Experience

The Caffevolve™ device serves two types of users: **Resident Users**, with stored profiles and personalised blend recommendations, and **Guest Users**, who brew through a quick-profile or express path without a permanent record. This section walks through every screen in sequence — device setup, profile creation, brew paths, Guest paths, and External Consumption.

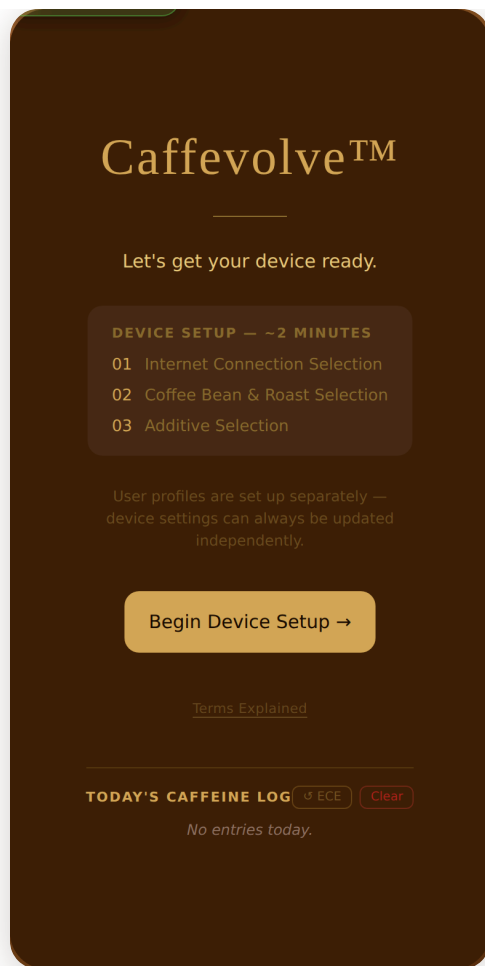
The Brew Session

A Brew Session is the complete sequence from device wake to a finished cup, and it comprises two phases. Phase 1 of the Brew Session — Pre-Brew Configuration — begins at the Brew Gateway, the device's brew entry point, and ends at hold-to-brew confirmation: the user selects a profile, declares tonight's bedtime, makes the brew-time selections, and reviews the caffeine recommendation, with the option to adjust additives or caffeine level. Every residual calculation, dual-ceiling evaluation, and blend-ratio computation happens in this phase; no physical machine operation has yet begun. Phase 2 of the Brew Session — the Active Brew Cycle — is initiated by hold-to-brew confirmation and carries out every physical operation in sequence: gravimetric bean dispensing, blending, grinding, brewing, and synchronized additive dispensing. No profile changes or recalculations occur once the Active Brew Cycle has begun.

These Brew Session phases are distinct from the AFS enrollment phases of Section 3 — Orientation, Introduction, and Calibration — which describe the feedback system's rollout across weeks of use, not the two stages of a single brew.

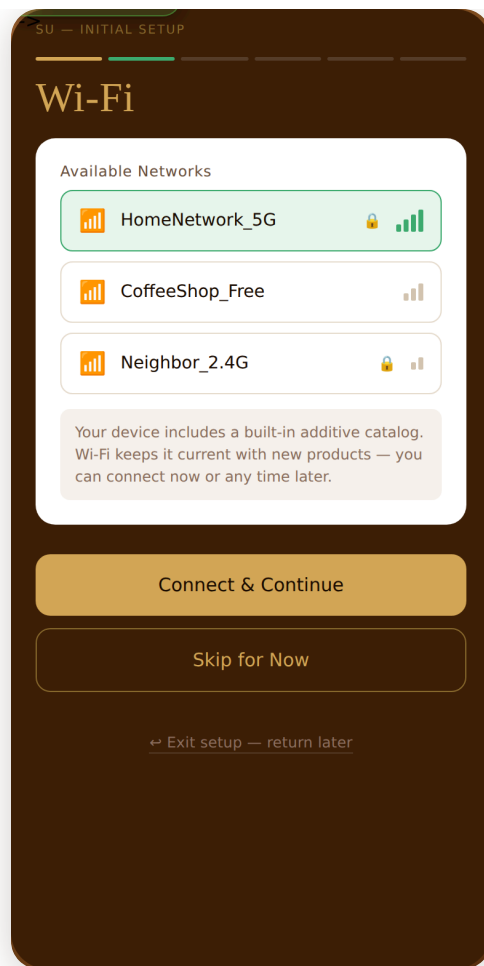
Device Set-up

SU0 · SU2 · SU-BEAN · SU-BEAN-TBP · SU5 · SU4a · SU5a



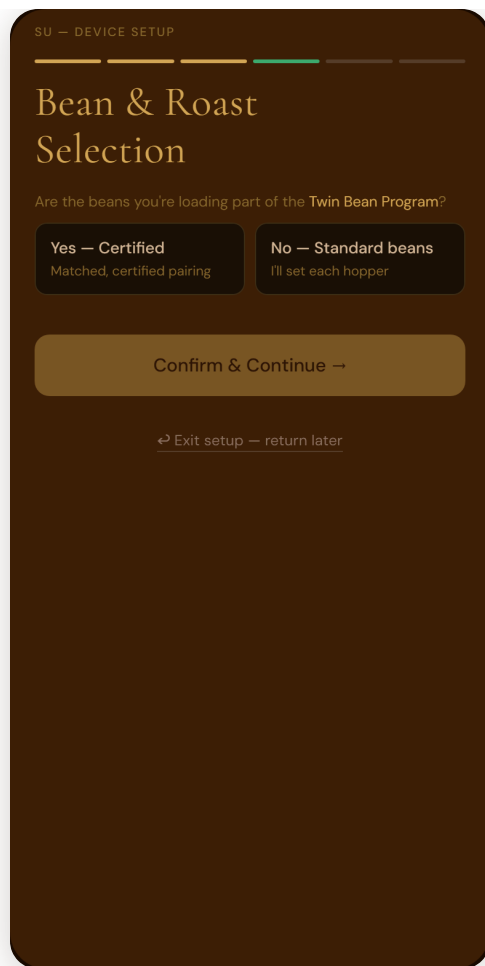
SU0 — Device Welcome

SU0 — Device Welcome. Lists the three setup steps: *01 Internet Connection Selection*, *02 Coffee Bean & Roast Selection*, *03 Additive Selection*. User profiles are set up separately. Tap *Begin Device Setup* to start.



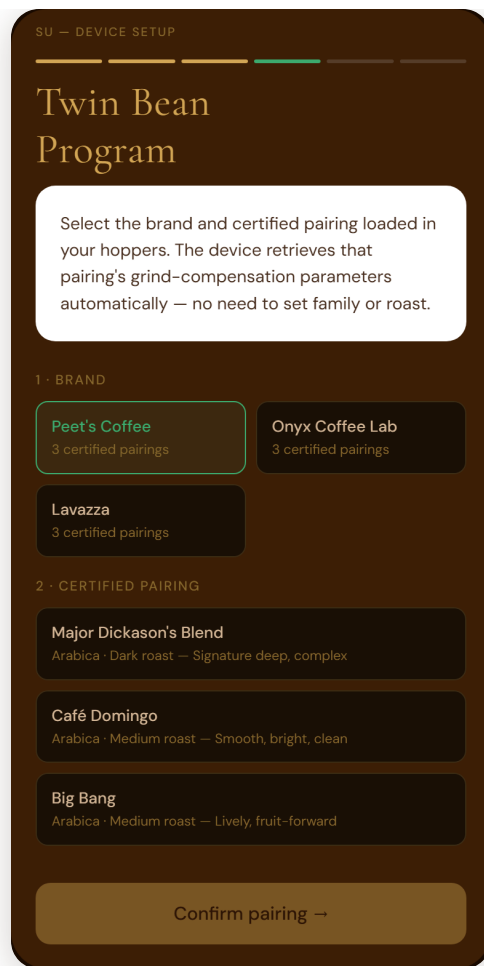
SU2 — Internet Connection Selection

SU2 — Internet Connection Selection. Available Wi-Fi networks are listed. The device ships with a built-in additive catalog — Wi-Fi keeps it current with new products and seasonal additions. The user selects a network, or taps *Skip for Now* to connect later.



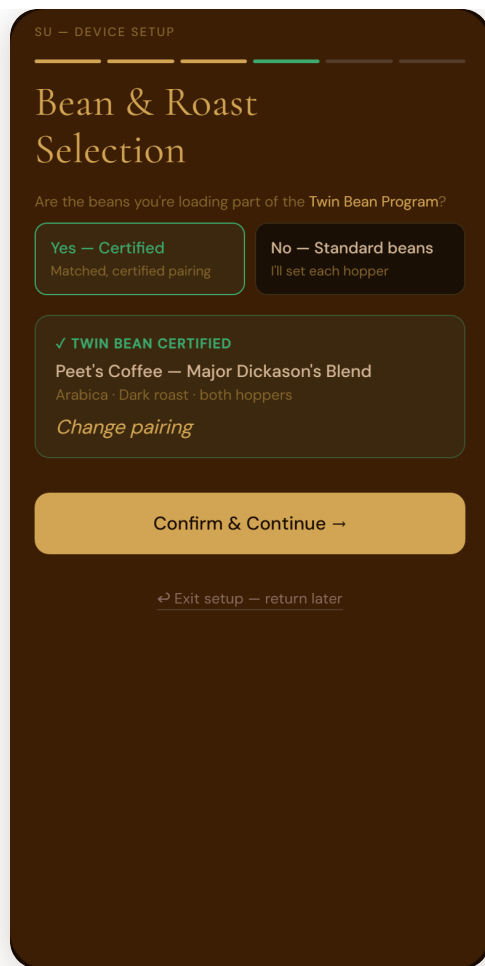
SU-BEAN — Bean & Roast Selection

SU-BEAN — Bean & Roast Selection. Setup step 02. The device first asks whether the beans being loaded are part of the *Twin Bean Program* — Caffevolve’s certified-pairing program, in which a partner roaster supplies a matched caffeinated and decaffeinated pair. The answer routes the user down one of two paths: *Yes — Certified* or *No — Standard beans*.



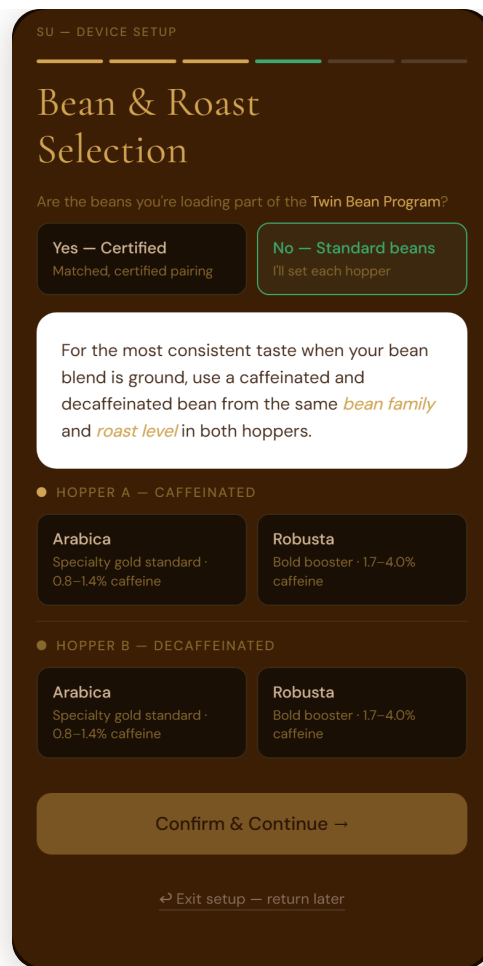
SU-BEAN-TBP — Twin Bean Program

SU-BEAN-TBP — Twin Bean Program. The certified path. The user selects the partner brand and the specific certified pairing loaded in the hoppers. The device retrieves that pairing’s bean family, roast level, and caffeine density automatically from the certification record — so no hopper values need to be entered by hand. Shown: Peet’s Coffee, with three certified pairings available.



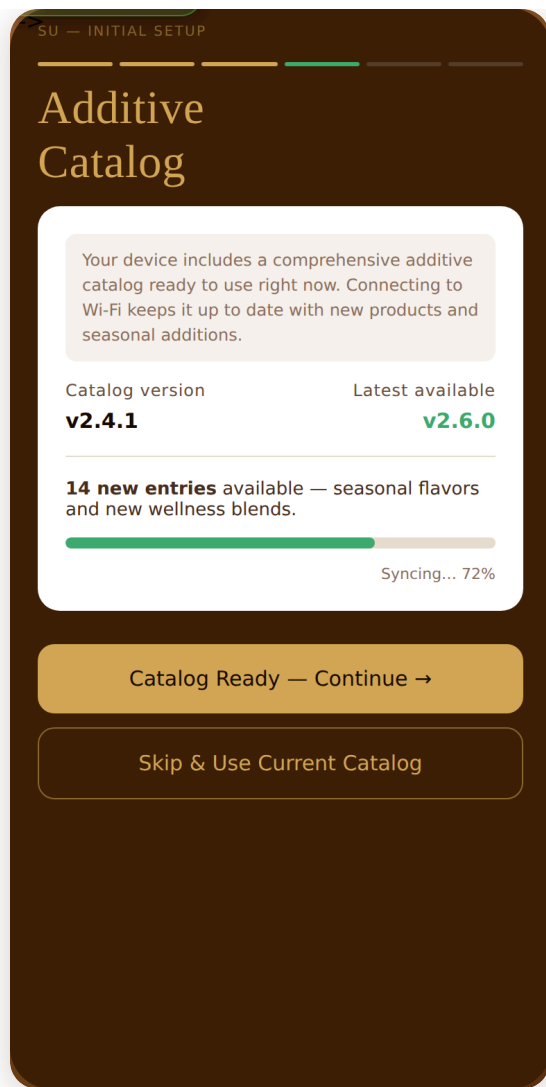
SU-BEAN — Twin Bean Certified

SU-BEAN — Twin Bean Certified. Returning from the pairing picker, the selection is confirmed inline: *Peet's Coffee — Major Dickason's Blend, Arabica · Dark roast · both hoppers*. A *Change pairing* link reopens the picker, and *Confirm & Continue* is now enabled.



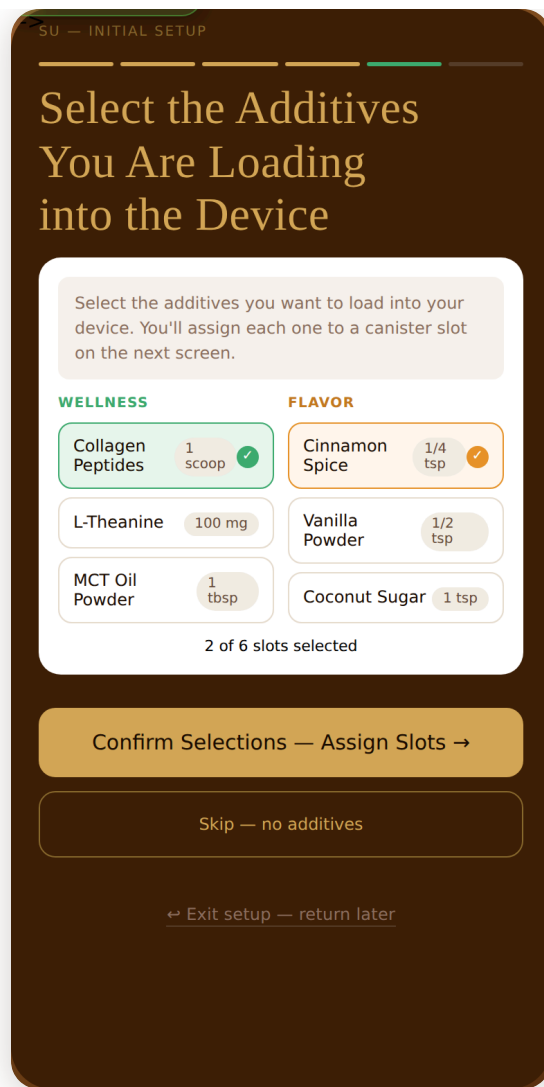
SU-BEAN — Standard Bean Path

SU-BEAN — Standard Bean Path. Choosing *No — Standard beans* reveals manual hopper configuration. Guidance recommends a caffeinated and decaffeinated bean from the same *bean family* and *roast level* in both hoppers for the most consistent grind. The user configures *Hopper A — Caffeinated* and *Hopper B — Decaffeinated* independently, selecting Arabica or Robusta and then a roast level for each.



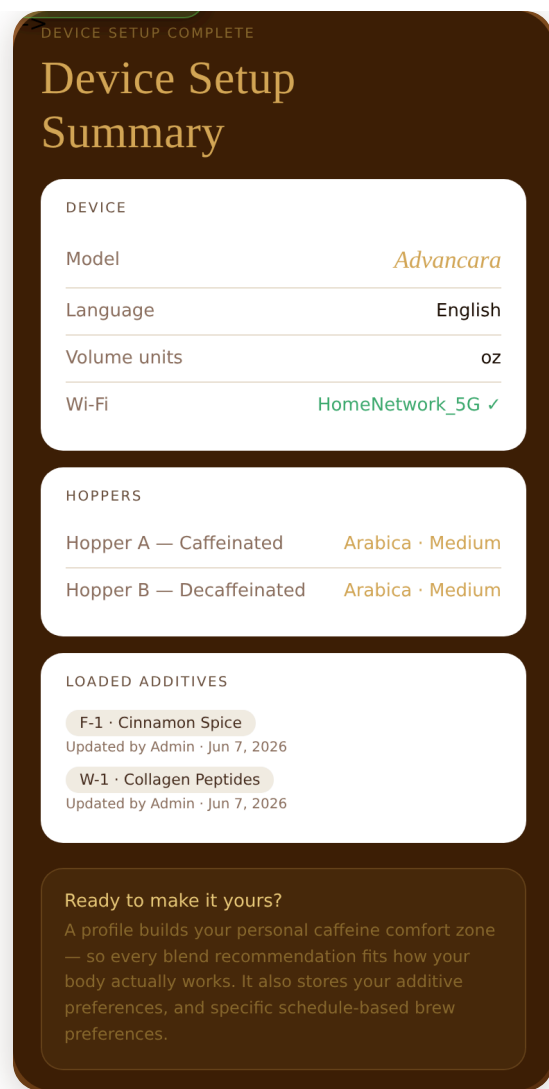
SU5 — Additive Catalog

SU5 — Additive Catalog. Once Wi-Fi is connected the device syncs its catalog. Shown: current version v2.4.1, latest v2.6.0, with 14 new seasonal flavors and wellness blends syncing at 72%. Continue once ready, or skip and use the current catalog.



SU4a — Additive Loading

SU4a — Select the Additives You Are Loading into the Device. The user selects additives across Wellness and Flavor categories, each showing its default dose. Selections are confirmed and assigned to canister slots on the next screen. Two of six slots selected here.



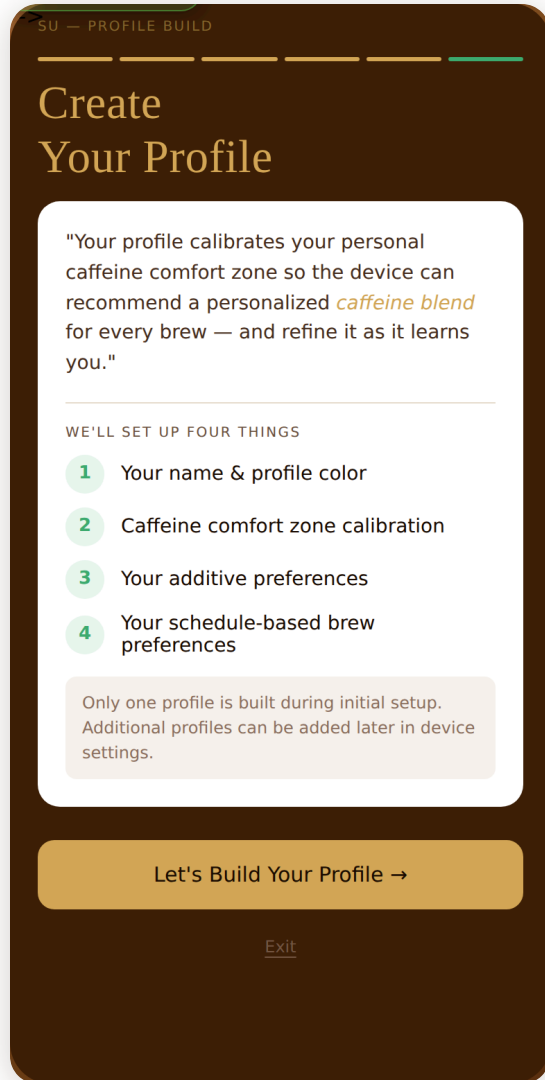
SU5a — Device Setup Summary

SU5a — Device Setup Summary. Full summary of completed device configuration: Device (model Advancara, English, oz, Wi-Fi connected), Hoppers (Hopper A — Caffeinated: Arabica · Medium; Hopper B — Decaffeinated: Arabica · Medium), and Loaded Additives (F-1 · Cinnamon Spice; W-1 · Collagen Peptides). A prompt invites the user to create a Resident User profile.

Resident User — User Profile Set-up

A Resident User is a regular user of the device with a stored profile holding their calibrated Caffeine Comfort Zone, brew preferences, additive selections, and grind setting.

SU6 · SU7 · SU8 · SU8-Confirm · SU9 · SU12 · SU-GRIND · SU13



SU6 — Create Your Profile

SU6 — Create Your Profile. Introduces four setup steps: *1 Your name & profile color*, *2 Caffeine Comfort Zone calibration*, *3 Your additive preferences*, *4 Your schedule-based brew preferences*. Only one profile is built during initial setup; additional profiles can be added later.

SU7 — Your Name & Color

SU7 — Your Name & Color. Profile Step 1 of 4. The user enters their first name and initials (auto-generated from the name, editable to 2–3 characters) and chooses a profile color. The initials circle previews the chosen color. Name truncates at approximately 8 characters on the profile carousel.

SU8 — Tell Us About Your Daily Coffee Routine

SU8 — Tell Us About Your Daily Coffee Routine. Profile Step 2 of 4. An 8-question sensitivity calibration instrument, presented one question at a time to establish initial sensitivity parameters for the user. Over the next few weeks the device presents outcome-related questions (FR System) to the user to refine its understanding of the user's sensitivity and adjusts the sensitivity parameters if needed.

SU — PROFILE STEP 2 OF 4

Review Your Answers

Here's what you told us. Tap any row to go back and change it — or tap **Looks Right** to compute your comfort zone.

Daily volume	3 drink(s) edit
Wake-up drink	A regular cup of coffee edit
Energize threshold	Two edit
Too-much point	~Three edit
Consistency	Pretty consistent edit
Last-drink cutoff	15:00 edit
Usual bedtime	22:30 edit
Late-caffeine sleep	A little harder to fall asleep edit

← Edit

Looks Right →

SU8-Confirm — Review Your Answers

SU8-Confirm — Review Your Answers. All eight instrument answers displayed together: Daily volume, Wake-up drink, Energize threshold, Too-much point, Consistency, Last-drink cutoff, Usual bedtime, and Late-caffeine sleep effect. Tap any row to edit, or tap *Looks Right* to compute the Caffeine Comfort Zone.

SU — PROFILE STEP 2 OF 4

Your Comfort Zone

Moderate sensitivity

Your answers suggest a **moderate caffeine sensitivity** — you handle a typical amount comfortably, and your body clears most of it before sleep. These are your starting parameters.

PEAK-ZONE FLOOR 48 mg from your answers	WAKE-HOURS CEILING 285 mg from your answers
BEDTIME CEILING 50 mg conservative prior, contrast-adjusted	BURN RATE · HALF-LIFE 6 hr back-solved from cutoff & bedtime

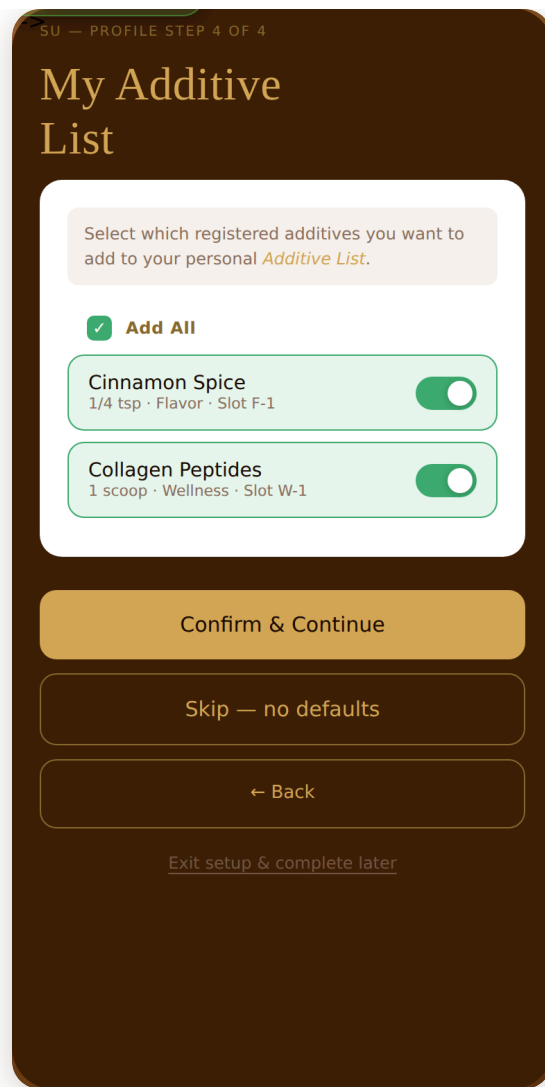
These are starting points — want to adjust the ceilings manually?

Fine-tune my ceilings manually

Accept & Continue →

SU9 — Your Caffeine Comfort Zone

SU9 — Your Caffeine Comfort Zone. Four computed parameters: *Peak Zone Floor 48 mg* (from answers), *Wake Hours Ceiling 285 mg* (from answers), *Bedtime Ceiling 50 mg* (CBI-derived from declared cutoff & bedtime, conservatively weighted), *Caffeine Decay Rate 6 hr* (back-solved from cutoff & bedtime). Accept and continue, or fine-tune ceilings manually.



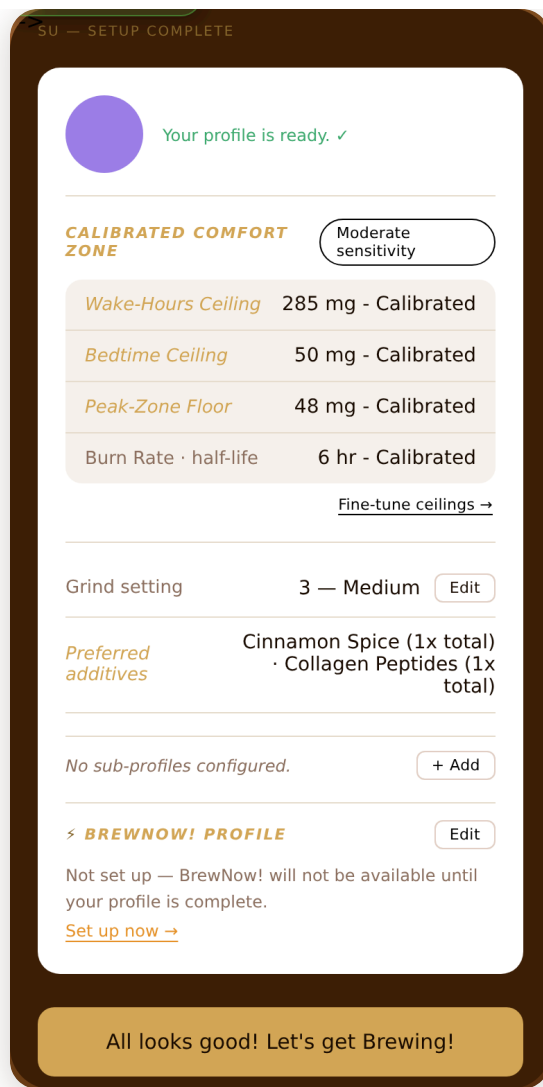
SU12 — My Additive List

SU12 — My Additive List. Profile Step 3 of 4. The user selects which device-registered additives to include in their personal Additive List. Shown: Cinnamon Spice and Collagen Peptides have been selected. Only these two additive choices will appear in the brew-time carousel for the user.



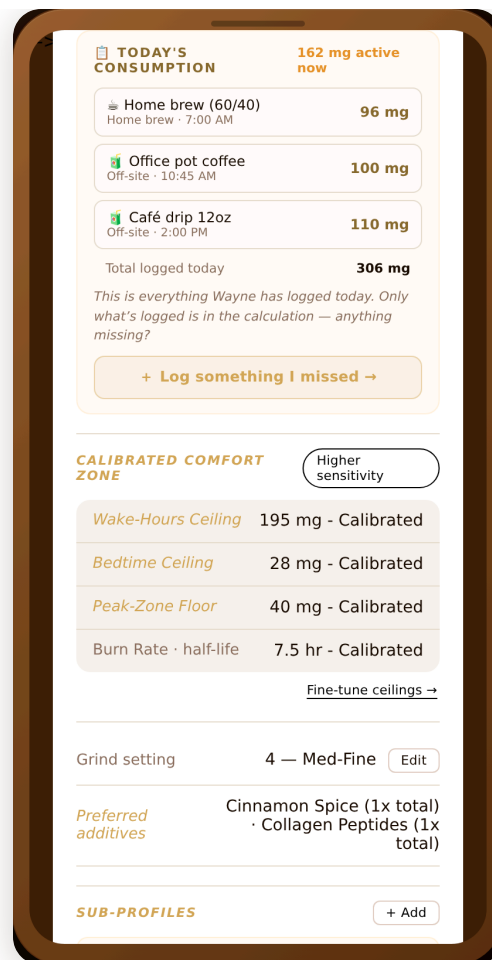
SU-GRIND — Grind Setting

SU-GRIND — Grind Setting. Six settings from Coarse to Extra Fine, each described by taste impact: *1 Coarse — light, bright, clean; 2 Med-Coarse — smooth and mellow; 3 Medium — balanced and approachable; 4 Med-Fine — fuller body, richer flavor; 5 Fine — intense and concentrated; 6 Extra Fine — maximum intensity, dense, syrupy.*



SU13 — Profile Summary

SU13 — Profile Summary. Setup complete. Example displays: Calibrated Caffeine Comfort Zone (a moderate-sensitivity profile, shown for illustration — WHC 285 mg, Bedtime Ceiling 50 mg, Peak Zone Floor 48 mg, Caffeine Decay Rate 6 hr, all Calibrated), Grind Setting 3 — Medium, Preferred additives (Cinnamon Spice, Collagen Peptides). Sub-profile status and BrewNow! Profile status (not set up). This is covered in the following section. When the Profile Summary is opened later to review an established profile — by selecting the profile and choosing *Review my Profile* — it additionally presents a **Today's Consumption** summary and a **My Workplace Drinks** section, both described below. Tap *All looks good! Let's get Brewing!*



SU13 — Today's Consumption

Today's Consumption. Reached by selecting a profile and choosing *Review my Profile*, the Profile Summary presents a live, per-profile account of the day: each caffeine event logged since midnight — device brews, off-site drinks logged through Caffevolve Go, and any residual carried forward from the prior day — each shown with its time, type, and caffeine amount, together with a running total and the residual caffeine level active now. The summary reflects only what the user has logged; it does not certify that the day is complete. It is therefore framed as a confirmation — an *anything missing?* prompt with a one-tap path to log an omission — so the user can attest that the day is fully accounted for before a late-day decision to brew or purchase.

Optional Profile Additions — BrewNow! Profile Set-up · SU12c

SU — BREWNOW! PROFILE

BrewNow! Profile

These settings are used when you tap BrewNow! outside of any sub-profile window. Your blend is always calculated fresh by the residual model.

Serving size

8 oz

12 oz

16 oz

20 oz

Serving count

−

1

+

Brew strength

Light

Medium

Strong

Your preferred *BrewNow! Additives*

Toggle on which additives to include in **BrewNow!** brews outside of sub-profile windows.*

Cinnamon Spice

1/4 tsp · Flavor

−

1.0x

+

☐

Collagen Peptides

1 scoop · Wellness

−

1.0x

+

☐

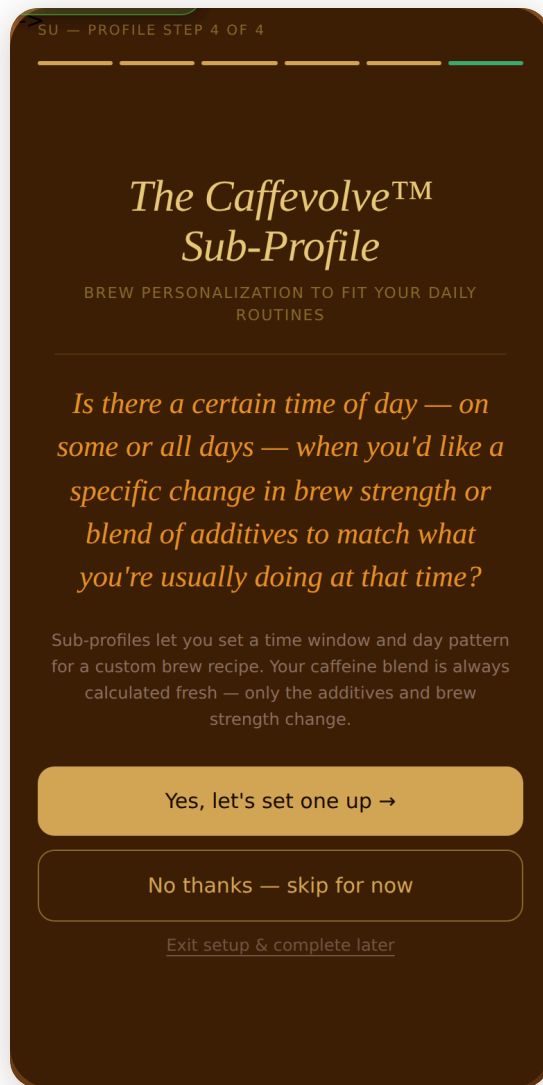
* When you tap BrewNow!, the device checks the current time against your sub-profile windows. If a window is active, its additive selections apply. Outside of any window, these BrewNow! defaults are used instead.

SU12c — BrewNow! Profile

SU12c BrewNow! Profile. Configures express brew defaults applied when the user chooses the BrewNow! path. Profile includes; serving size (12 oz selected), serving count, brew strength, and additives to add to the brew automatically — Cinnamon Spice and Collagen Peptides were chosen in the example shown. The recommended caffeine blend is never a user profile element and is always calculated at the time of the brew.

Optional Profile Additions — Sub-Profile Set-up · SU12a · SU12b · SU12b-CONFIRM

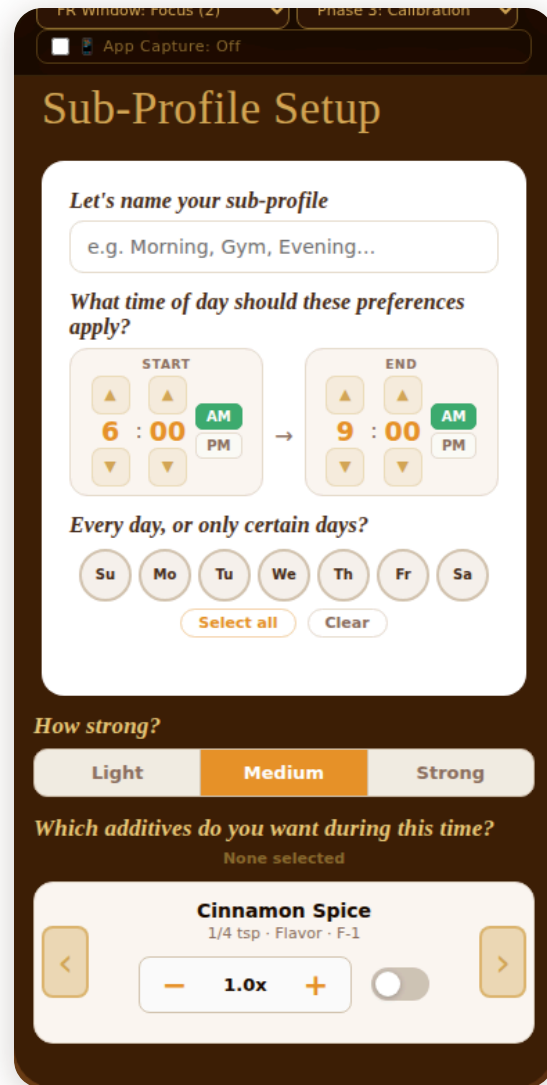
Sub-profiles define a user-chosen time window and day pattern for a custom brew recipe. The caffeine blend is always calculated fresh — only additives and brew strength change. Up to 3 sub-profiles per profile on Originara and Culminara Office, and up to 6 on Advancara.



SU12a — The Caffevolve™ Sub-Profile

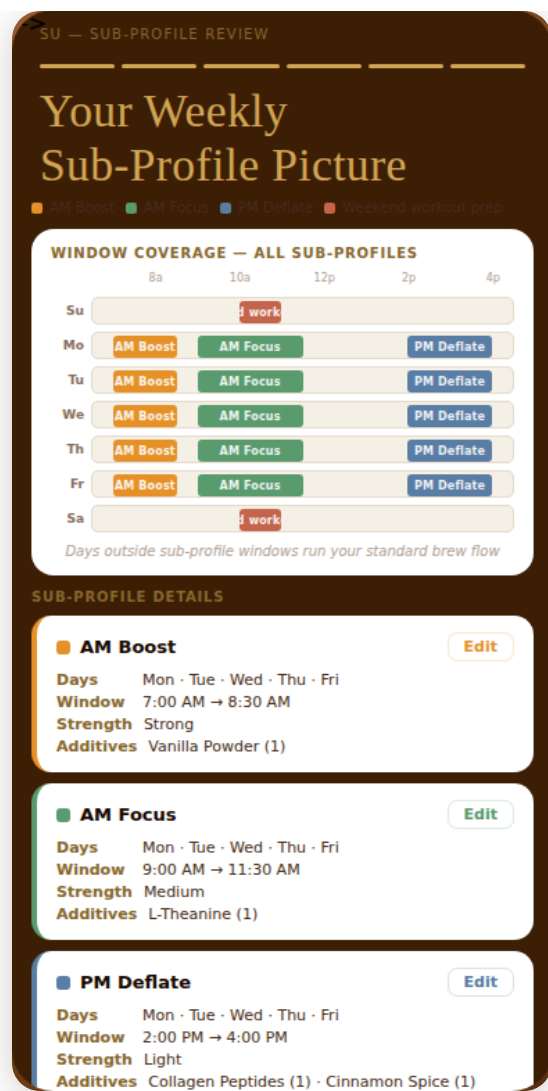
SU12a — The Caffevolve™ Sub-Profile.

Introduces the sub-profile concept with a question: *Is there a certain time of day — on some or all days — when you'd like a specific change in brew strength or blend of additives to match what you're usually doing at that time?* The user can proceed to set one up now, or skip and come back to it later.



SU12b — Sub-Profile Setup

SU12b — Sub-Profile Setup. Captures: a name (e.g. Morning, Gym, Evening), a time window (6:00 AM → 9:00 AM shown), and the day(s) it will apply on. User selects the brew strength and which additives to add to the brew during this window. Tap *See Complete Weekly Picture* to preview complete sub-profile coverage before confirming.

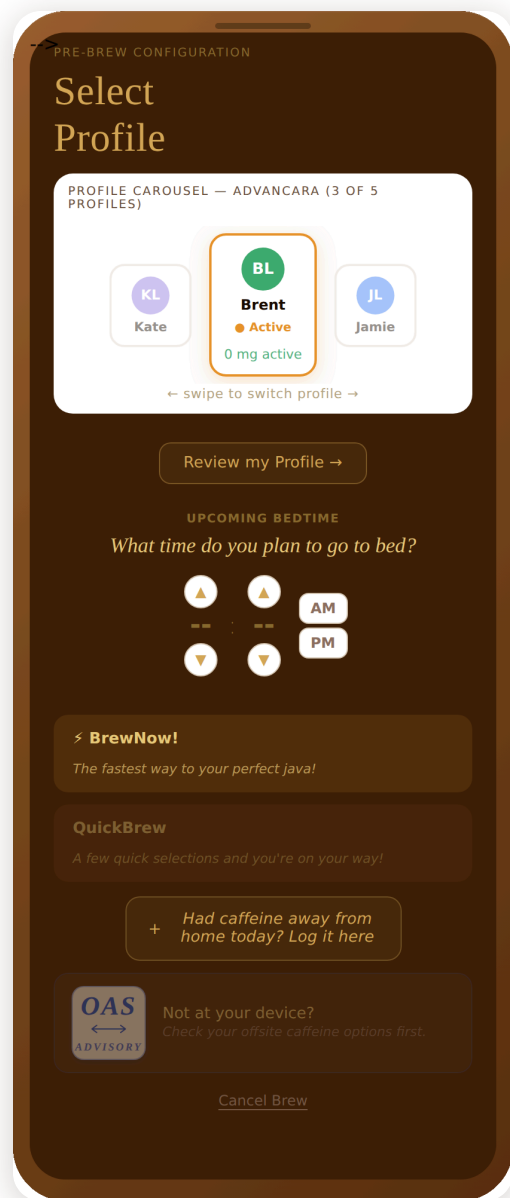


SU12b-CONFIRM — Your Weekly Sub-Profile Picture

SU12b-CONFIRM — Your Weekly Sub-Profile Picture. A calendar-style view showing window coverage across all sub-profiles. Days outside sub-profile windows run the standard brew flow. Confirm with *Confirm All Sub-Profiles* or return to add another.

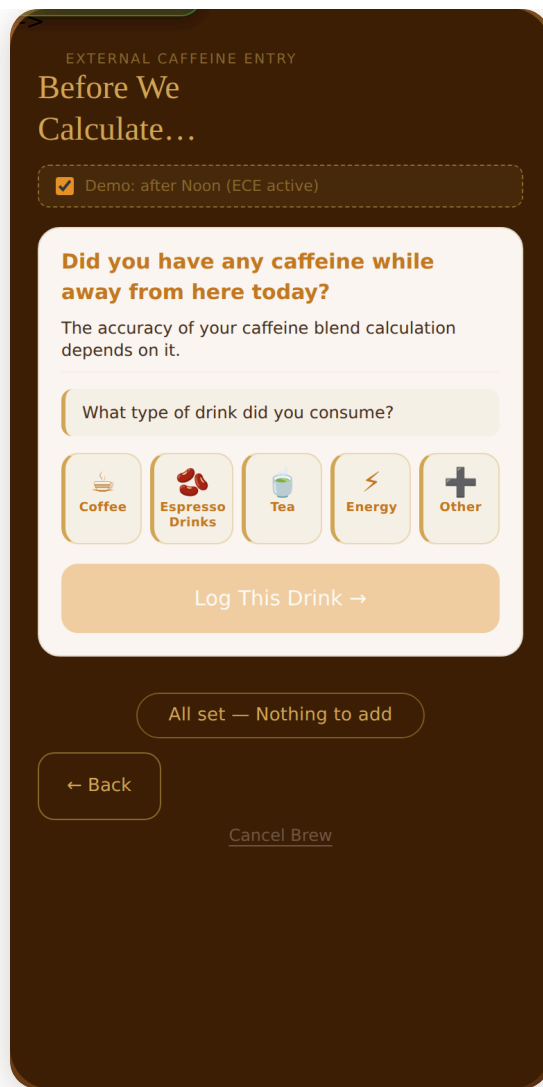
Resident User Brew Paths

BrewNow! — No Caffeine Adjustment · PS1a · ECE · BN-HTB



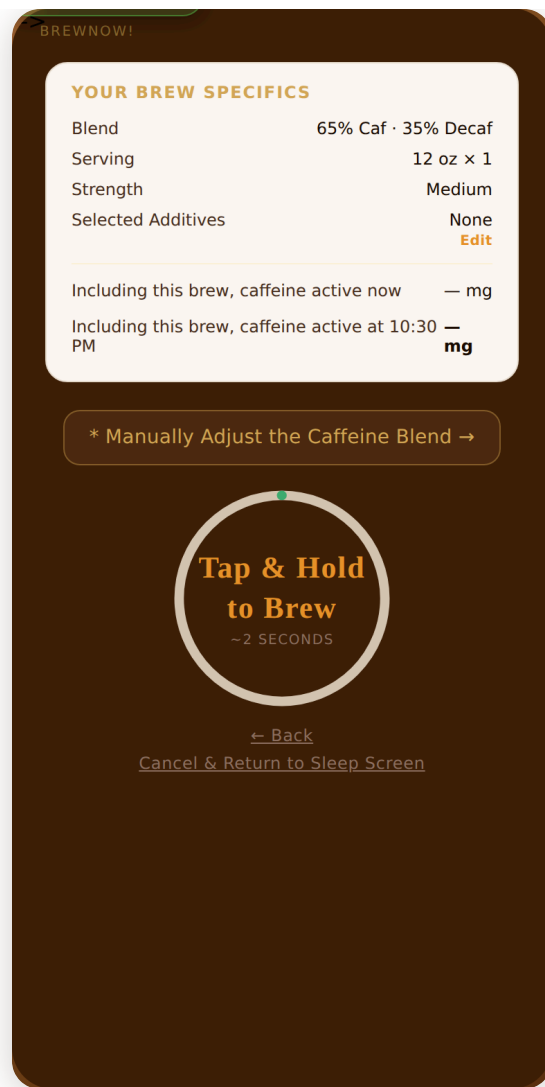
PS1a — Profile Selection

PS1a — Profile Selection. The Resident User selects their profile from the carousel, selects tonight’s bedtime, and chooses from five options moving forward. The first two options, BrewNow! and QuickBrew are device brews that occur at that time — the two remaining options, ECE and OAS are away from the device activities. Tapping the first option, BrewNow! initiates the express path to your desired drink.



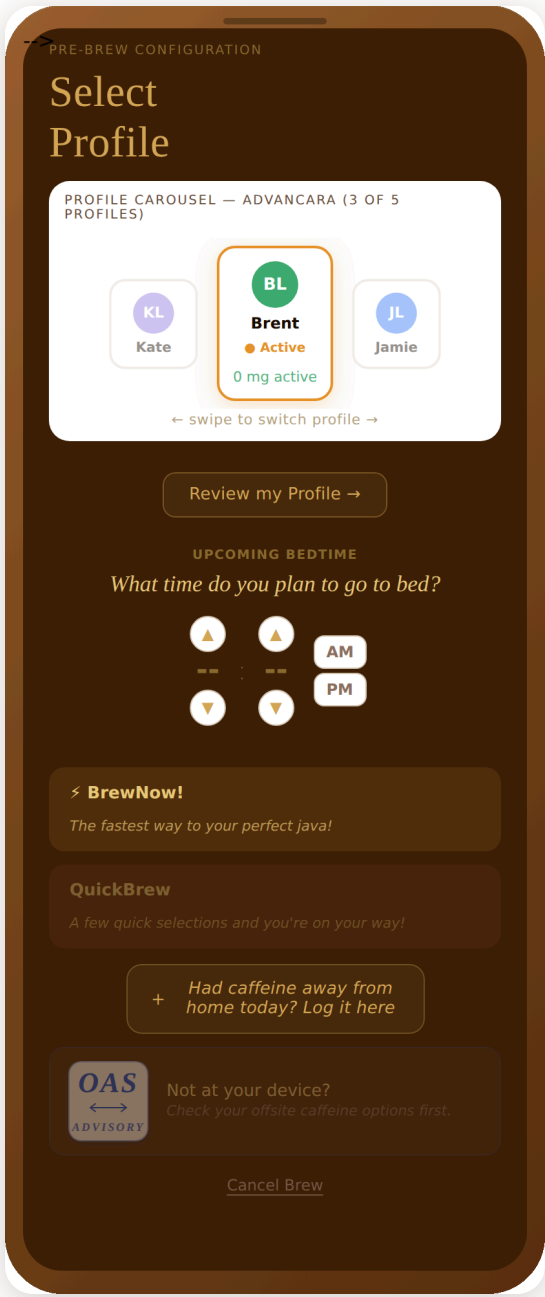
ECE — External Caffeine Entry

ECE — External Caffeine Entry. A couple of times each day, this ECE screen will surface automatically before a new brew can be initiated to ensure that all external caffeine consumption has been captured up to that point. This inquiry is necessary to ensure the caffeine recommendation for the next cup is based on accurate and up to date consumption information. If this does not apply, the user taps “All set - Nothing to add”. But if the inquiry does apply, the user logs the drink type, amount, and approximate time. The residual model is updated and all subsequent recommendations reflect the logged intake.



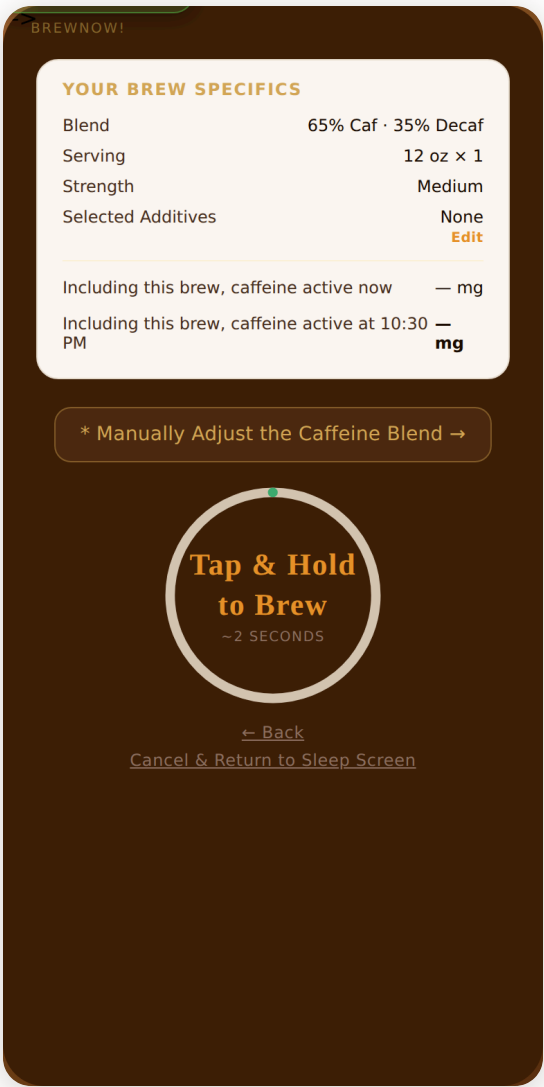
BN-HTB — Hold to Brew

BN-HTB — Hold to Brew. Once the user selects the BrewNow! option, a confirmation screen shows stored express defaults — size, strength, and preferred additives. The Blend recommendation is calculated once this option is chosen. Tap and Hold the brew button to initiate the brew cycle, or tap *Adjust manually* to open the blend adjustment bar.



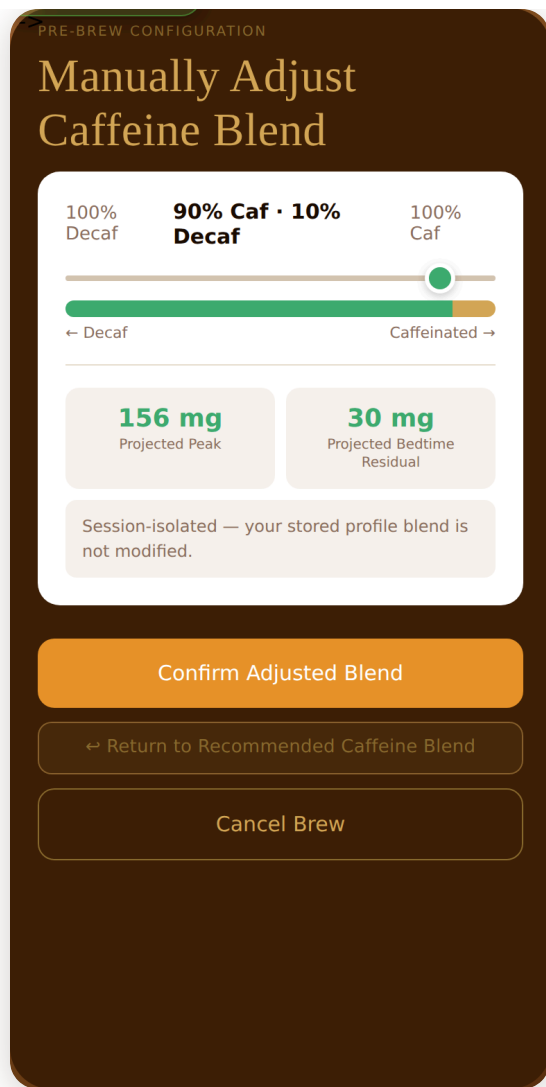
PS1a — Profile Selection

PS1a — Profile Selection. Profile selected, bedtime declared, BrewNow! initiated.



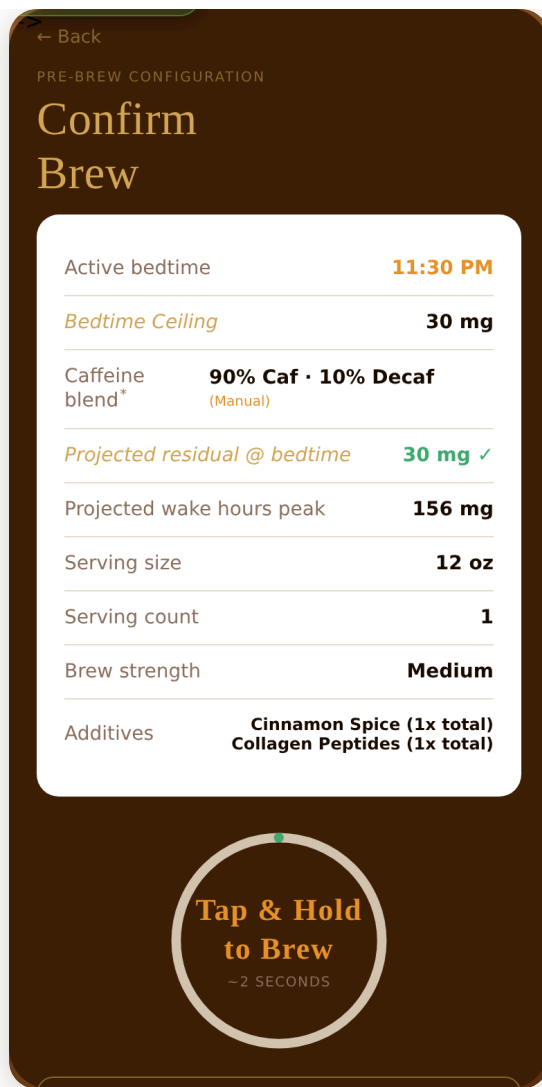
BN-HTB — Hold to Brew

BN-HTB — Hold to Brew. The user taps *Adjust manually* instead of holding to brew, opening the blend adjustment bar.



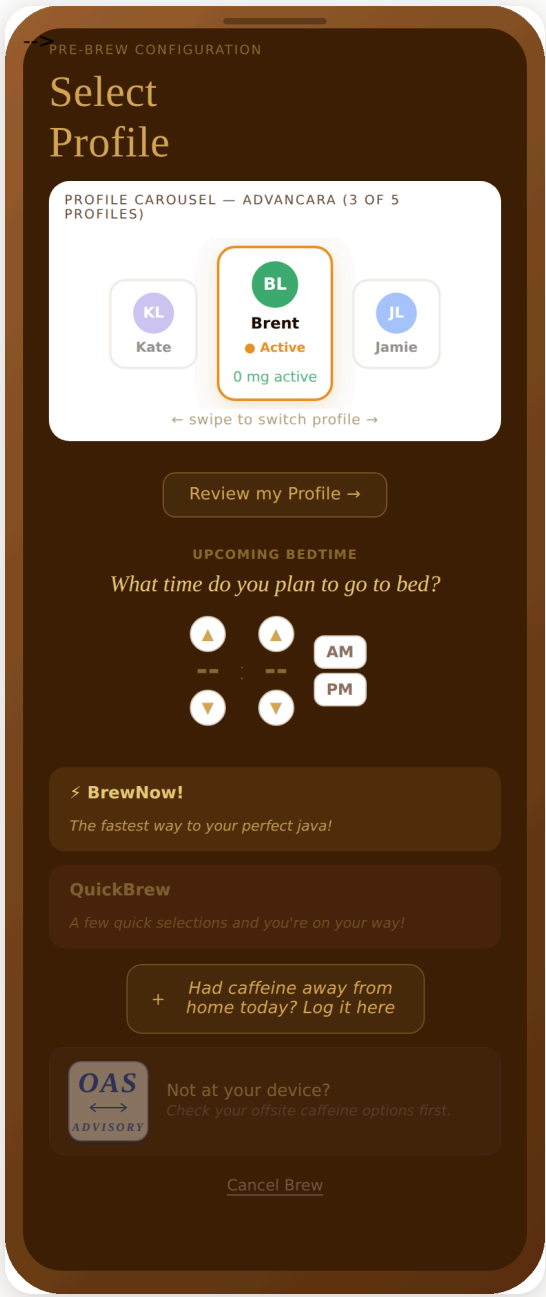
PS4 — Manual Caffeine Adjustment

PS4 — Manual Caffeine Adjustment. A live slider shifts the caffeine blend percentage. Projected Peak and Projected Bedtime Residual update in real time. Ceiling lines and the Peak Zone band are displayed. Ceiling violations trigger a warning banner.



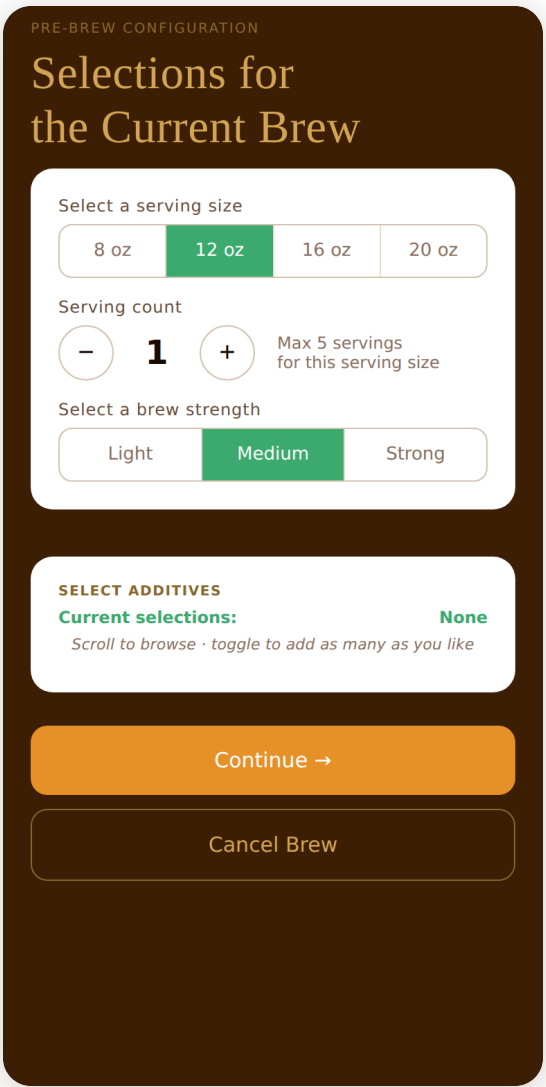
PS6-A — Confirm & Brew

PS6-A — Confirm & Brew. The adjusted brew confirmation screen. Displays the final blend and all parameters before the hold-to-brew gesture initiates the active brew cycle.



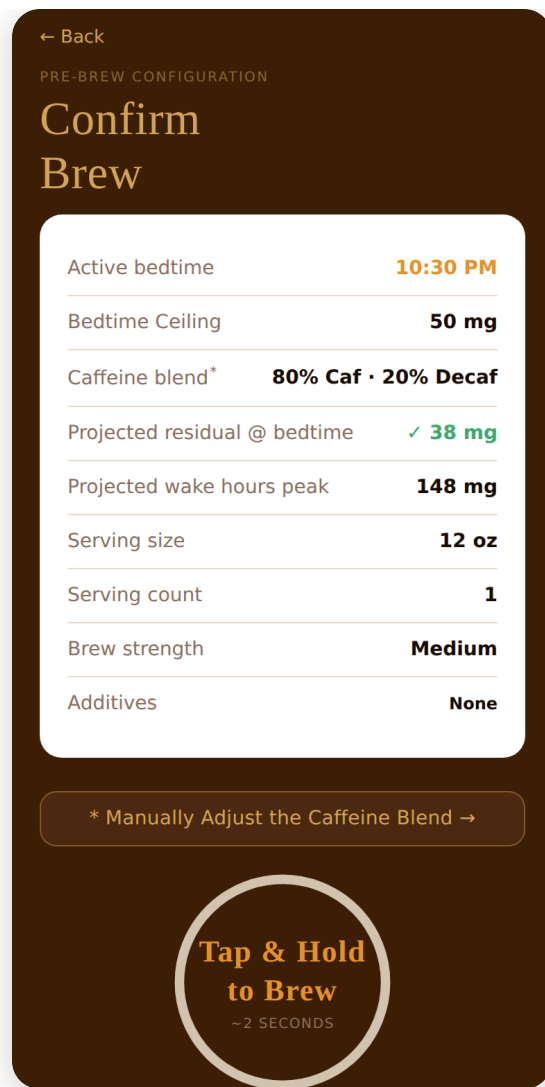
PS1a — Profile Selection

PS1a — Profile Selection. Profile selected, bedtime declared. The user taps QuickBrew to enter the full pre-brew configuration path.



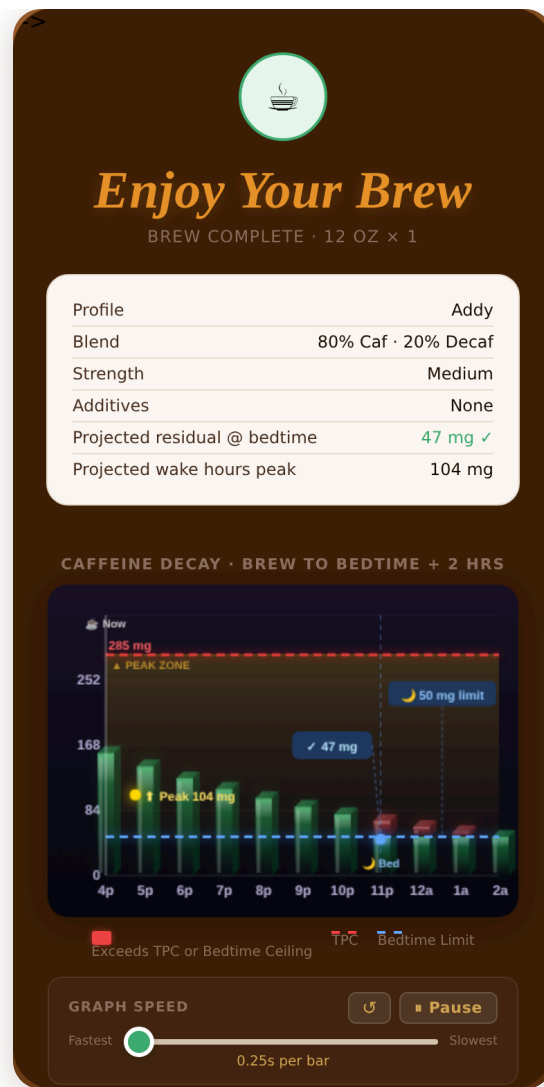
PS2a — Brew-Time Selections

PS2a — Brew-Time Selections. The user configures serving size and number of servings for this brew. If a sub-profile window is active, brew strength and additive selections are pre-populated from that sub-profile but remain editable; otherwise, brew strength and additive selections are made fresh at brew time, with additives selected from the user's Additive List. These parameters are always session-level — never stored to the profile.



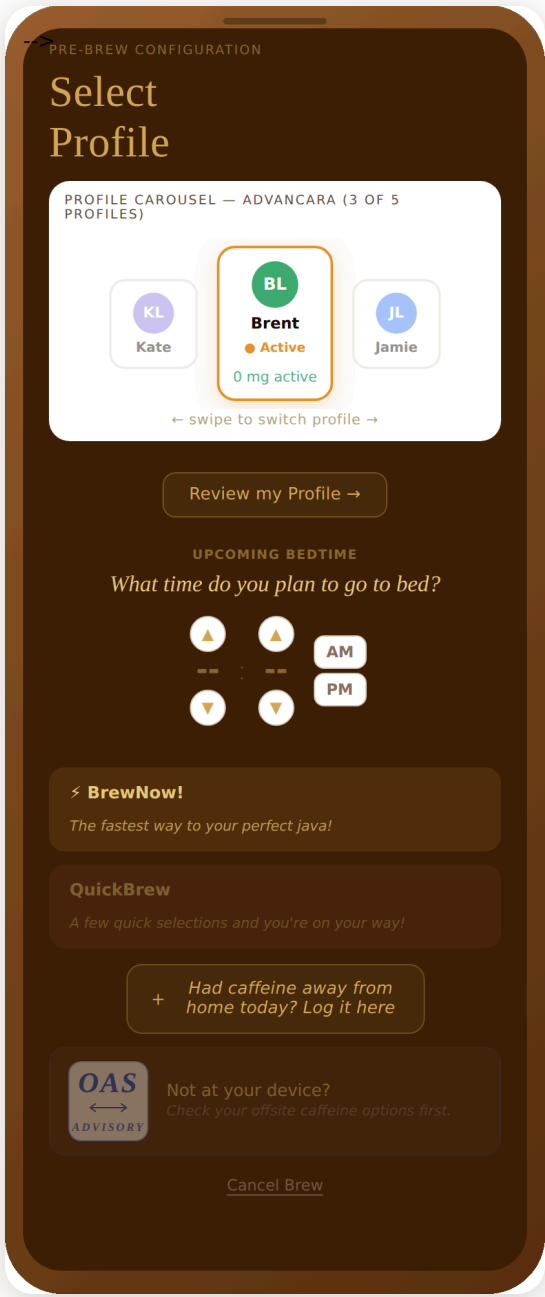
PS6-A — Recommendation & Brew

PS6-A — Recommendation & Brew. The caffeine blend percentage is presented alongside the Projected Peak and Projected Bedtime Residual values. The user accepts the recommendation and taps and holds to brew.



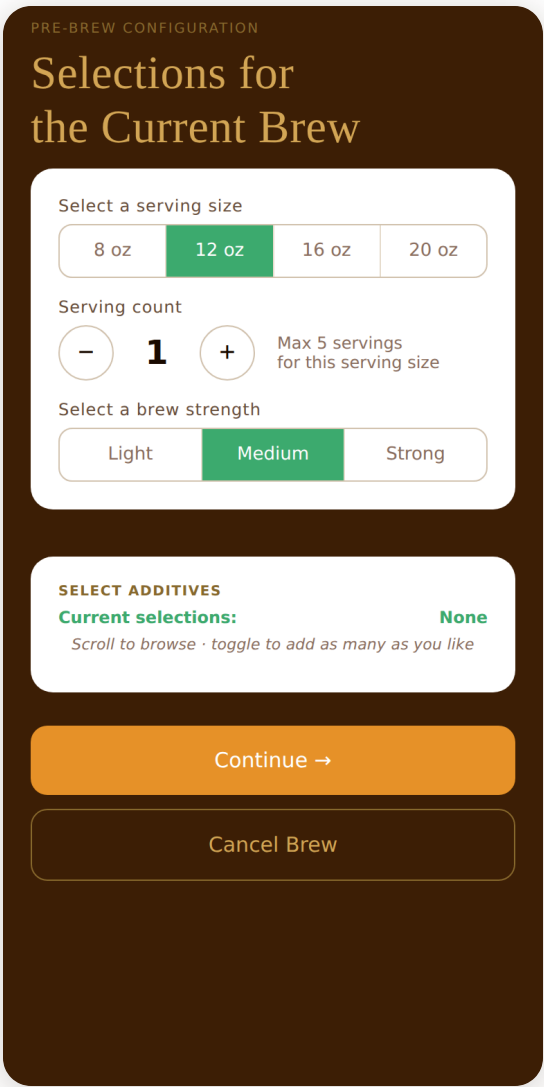
PS8 — Enjoy Your Brew

PS8 — Enjoy Your Brew. The brew complete screen. A session summary card confirms the profile, blend, strength, additives, projected residual caffeine at bedtime, and projected wake-hours peak for the brew just completed. A Caffeine Decay chart animates the projected caffeine curve from brew time through bedtime plus two hours, with reference lines marking the Wake Hours Ceiling and Bedtime Ceiling. The screen auto-advances to the home screen after a short countdown; the user may also tap *All set — Thank you!* to return immediately.



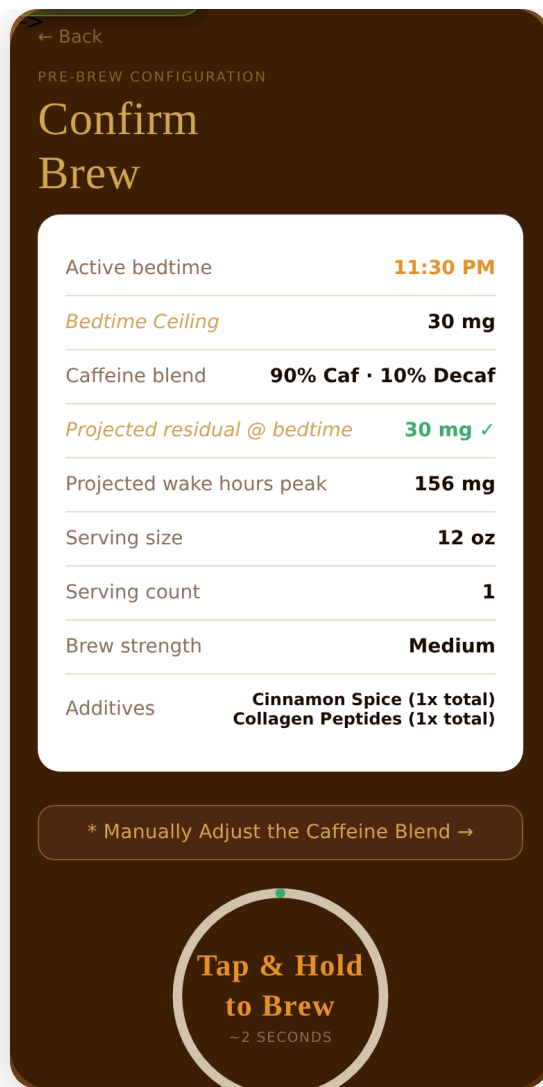
PS1a — Profile Selection

PS1a — Profile Selection. Profile selected, bedtime declared, QuickBrew initiated.



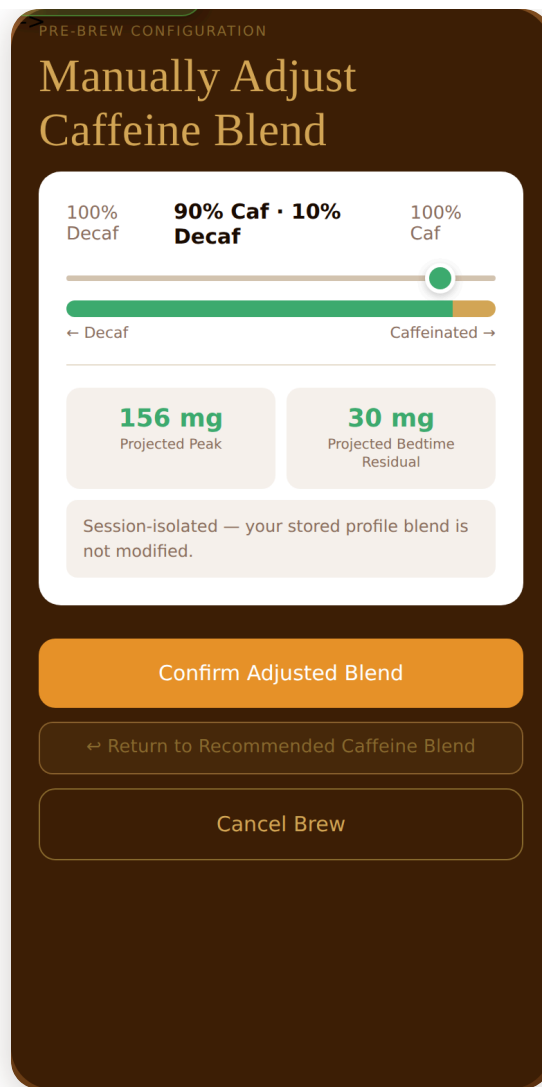
PS2a — Brew-Time Selections

PS2a — Brew-Time Selections. Size, strength, and additive selections for this brew.



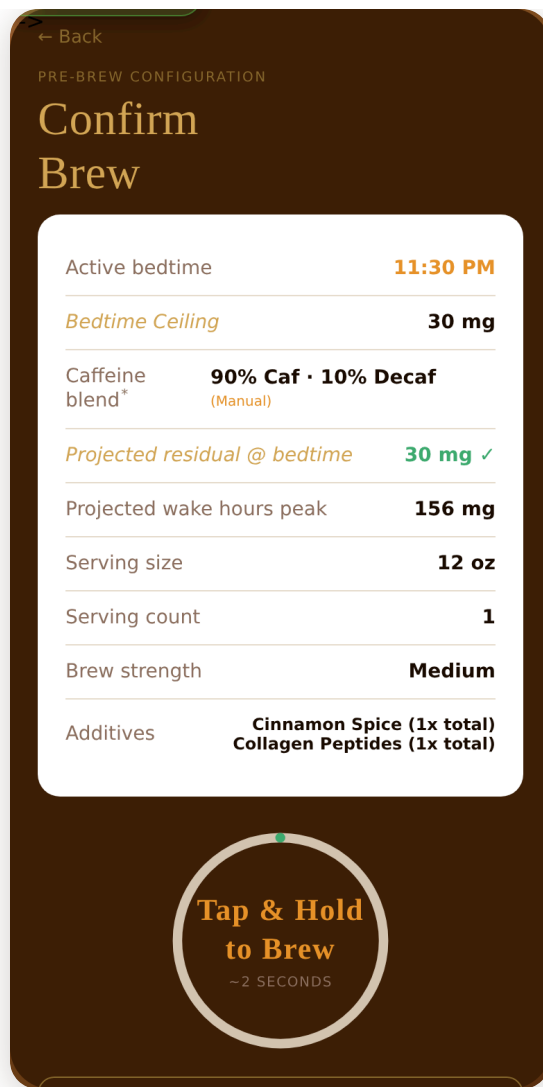
PS6-A — Recommendation

PS6-A — Recommendation. The system blend recommendation is presented. If the user chooses to not accept the recommendation, a tap on the *Adjust manually* rather than tap and hold to brew will allow the adjustment to proceed.



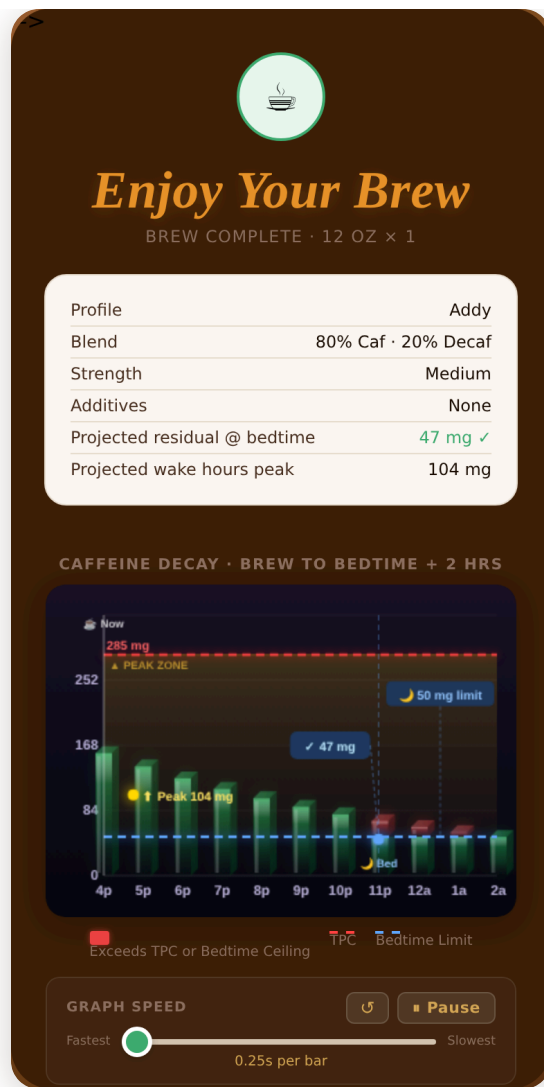
PS4 — Manual Caffeine Adjustment

PS4 — Manual Caffeine Adjustment. A live slider shifts the caffeine blend percentage. Projected Peak and Projected Bedtime Residual update in real time. Ceiling lines and the Peak Zone band are displayed. Ceiling violations trigger a warning banner.



PS6-A — Confirm & Brew

PS6-A — Confirm & Brew. Final confirmation with the adjusted blend. Hold to brew initiates the active brew cycle.



PS8 — Enjoy Your Brew

PS8 — Enjoy Your Brew. The brew complete screen. A session summary card confirms the profile, blend, strength, additives, projected residual caffeine at bedtime, and projected wake-hours peak for the brew just completed. A Caffeine Decay chart animates the projected caffeine curve from brew time through bedtime plus two hours, with reference lines marking the Wake Hours Ceiling and Bedtime Ceiling. The screen auto-advances to the home screen after a short countdown; the user may also tap *All set — Thank you!* to return immediately.

Guest User

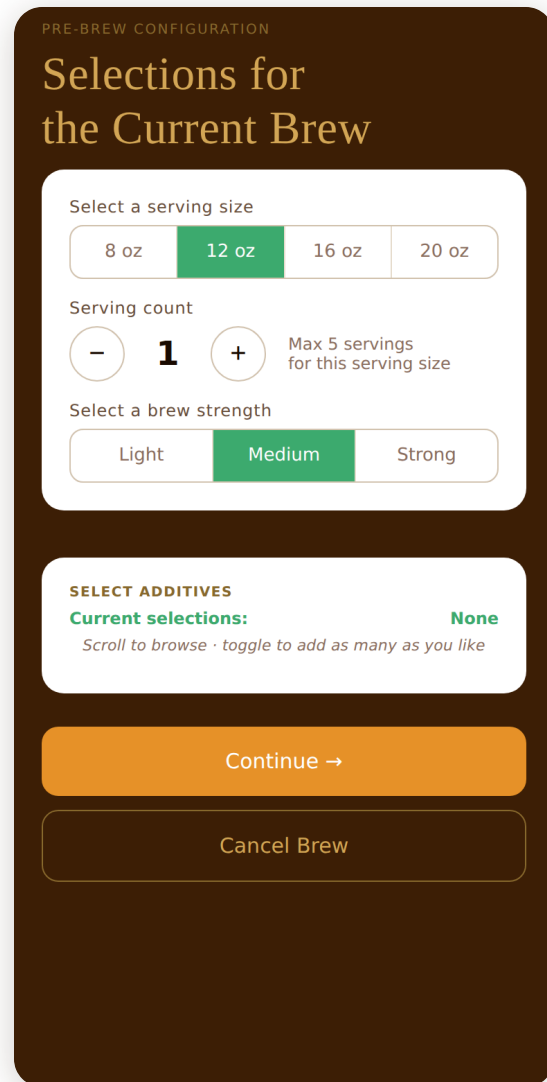
A guest brews without a stored Resident User Profile. Three paths are available for guests: **BrewNow!** for immediate manual-blend brewing, the **Quick Profile Path** for a personalized caffeine blend recommendation that establishes a Guest Profile, and the **Returning Guest Path** for guests whose Guest Profile was saved as a Returning Guest Profile within the past 72 hours.

Guest Brew Paths — BrewNow! · PS1b · PS2a · PS3a-C2



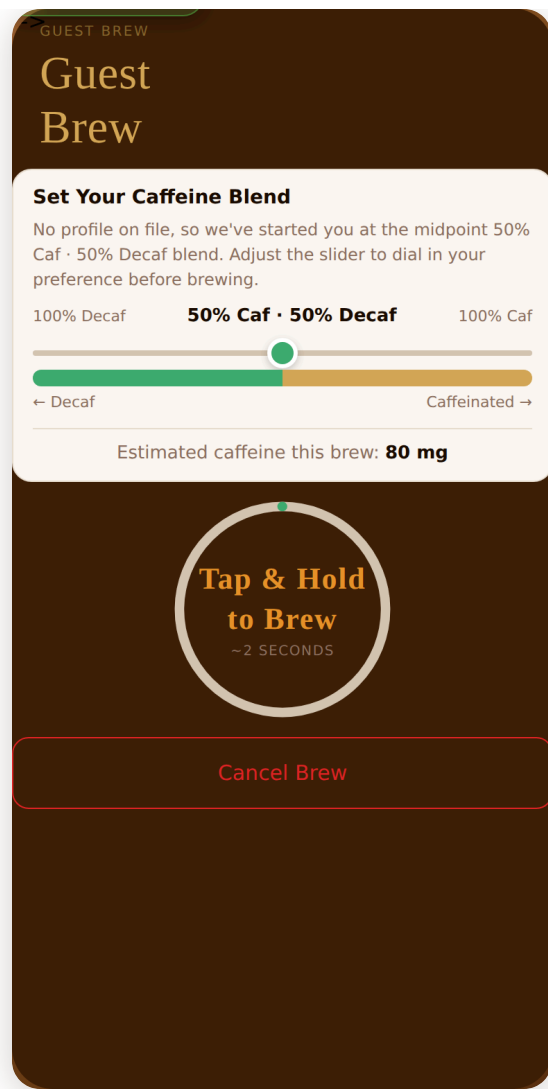
PS1b — Guest Entry

PS1b — Guest Entry. The guest entry point. Returning guests (within 72 hours) appear as cards with a direct ► Brew route. New guests tap either *BrewNow!* for the fastest path forward or *Quick Profile* for a limited scope profile with a complete caffeine blend recommendation.



PS2a — Brew-Time Selections

PS2a — Brew-Time Selections. Once the Guest taps *BrewNow!* on the entry screen, they are advanced to select the size, serving count, and strength of the brew. All device-loaded additives are available — no profile-level filtering applies.

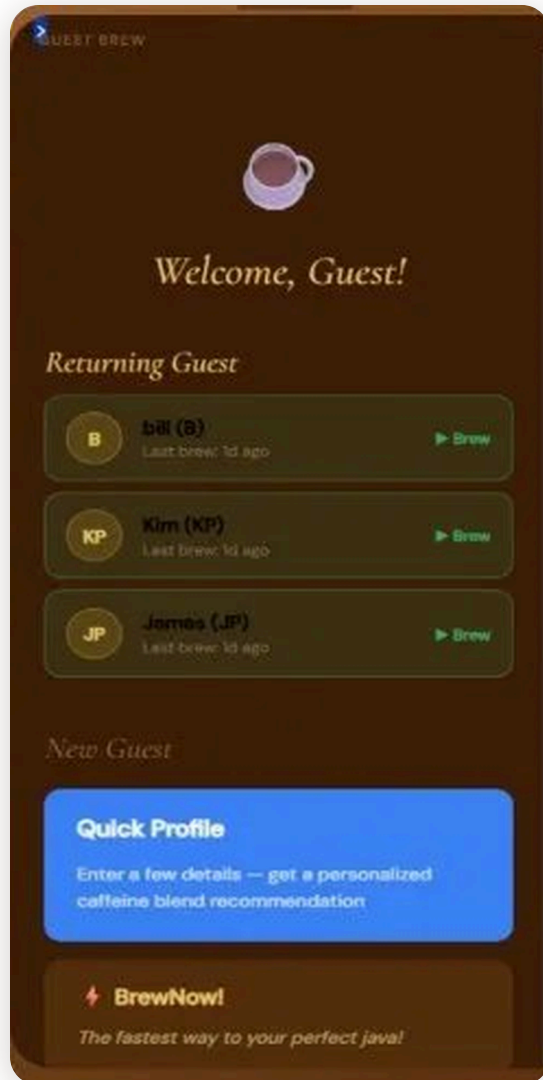


PS3a-C2 — Guest Blend

PS3a-C2 — Guest Blend. The guest manually sets their caffeine blend percentage with the slider. No recommendation is generated. Hold to brew confirms the selected blend and initiates the brew.

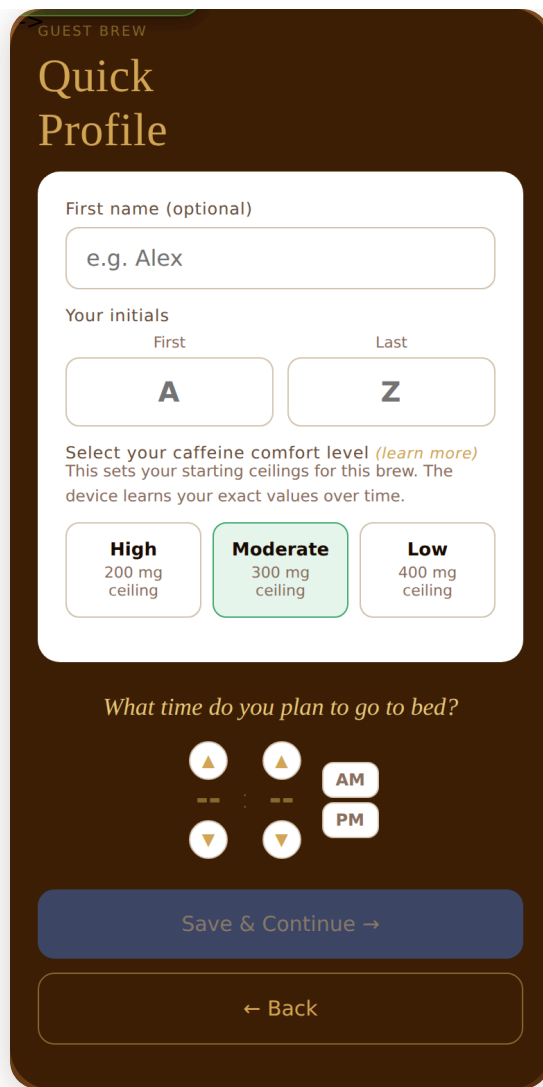
Guest Brew Paths — Quick Profile Path · PS1b · PS1c · ECE · PS2a · PS6-A

A new guest can receive a personalized blend recommendation via the Quick Profile Path, which establishes a Guest Profile. At the end of the flow the guest is asked whether they are likely to brew again: if so, the Guest Profile is saved as a Returning Guest Profile for reuse via the Returning Guest Path within the next 72 hours; if not, the Guest Profile is used for this brew only and then discarded.



PS1b — Guest Entry

PS1b — Guest Entry. The guest taps *Quick Profile* to begin the personalized guest brew path.



GUEST BREW

Quick Profile

First name (optional)

e.g. Alex

Your initials

First Last

A Z

Select your caffeine comfort level ([learn more](#))
This sets your starting ceilings for this brew. The device learns your exact values over time.

High
200 mg ceiling

Moderate
300 mg ceiling

Low
400 mg ceiling

What time do you plan to go to bed?

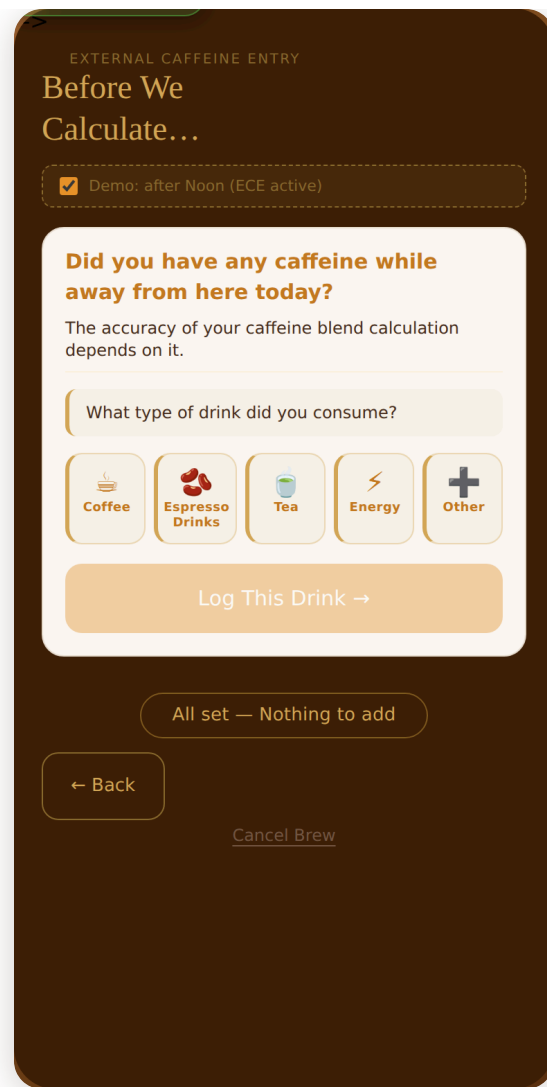
Time picker showing AM and PM options.

Save & Continue →

← Back

PS1c — Guest Quick Profile

PS1c — Guest Quick Profile. Name, initials, and caffeine sensitivity level (Higher / Moderate / Lower) are entered along with the planned bedtime, establishing the Guest Profile. On confirming, the guest is asked whether they are likely to brew again — if so, the Guest Profile is saved as a Returning Guest Profile for 72 hours; if not, it is used for this brew only and then discarded. As a Guest-only shortcut, the selected sensitivity tier also sets a default Caffeine Decay Rate — Higher = 7.0 hrs, Moderate = 6.0 hrs, Lower = 5.0 hrs — mirroring the same proportional logic as the sensitivity-based ceiling defaults — higher sensitivity pairs a lower ceiling default with a longer assumed half-life (7.0 hr), both the more conservative setting, so the two move in the same protective direction — rather than the CBI-derived personalized rate a Resident User Profile establishes. This tier-to-decay-rate link applies only to the Guest quick-profile default; for Resident



EXTERNAL CAFFEINE ENTRY

Before We Calculate...

☒ Demo: after Noon (ECE active)

Did you have any caffeine while away from here today?

The accuracy of your caffeine blend calculation depends on it.

What type of drink did you consume?

Coffee Espresso Drinks Tea Energy Other

Log This Drink →

All set — Nothing to add

← Back

[Cancel Brew](#)

ECE — External Caffeine Entry

ECE — External Caffeine Entry. This screen checks for unknown caffeine consumption before recommending the caffeine blend. Any unknown caffeine consumption that is captured here is factored into the blend recommendation.

Users, sensitivity (ceilings) and Caffeine Decay Rate remain independent traits, each derived from its own separate Caffeine Behavior Instrument question.

PRE-BREW CONFIGURATION

Selections for the Current Brew

Select a serving size

8 oz 12 oz 16 oz 20 oz

Serving count

- 1 + Max 5 servings for this serving size

Select a brew strength

Light Medium Strong

SELECT ADDITIVES

Current selections: None

Scroll to browse · toggle to add as many as you like

Continue →

Cancel Brew

PS2a — Brew-Time Selections

PS2a — Brew-Time Selections. Drink size, number of servings, brew strength, and additive selections are selected for this guest brew.

← Back

PRE-BREW CONFIGURATION

Confirm Brew

Active bedtime 11:30 PM

Bedtime Ceiling 30 mg

Caffeine blend 85% Caf · 15% Decaf

Projected residual @ bedtime 29 mg ✓

Projected wake hours peak 148 mg

Serving size 12 oz

Serving count 1

Brew strength Medium

Additives Cinnamon Spice (1x total)
Collagen Peptides (1x total)

* Manually Adjust the Caffeine Blend →

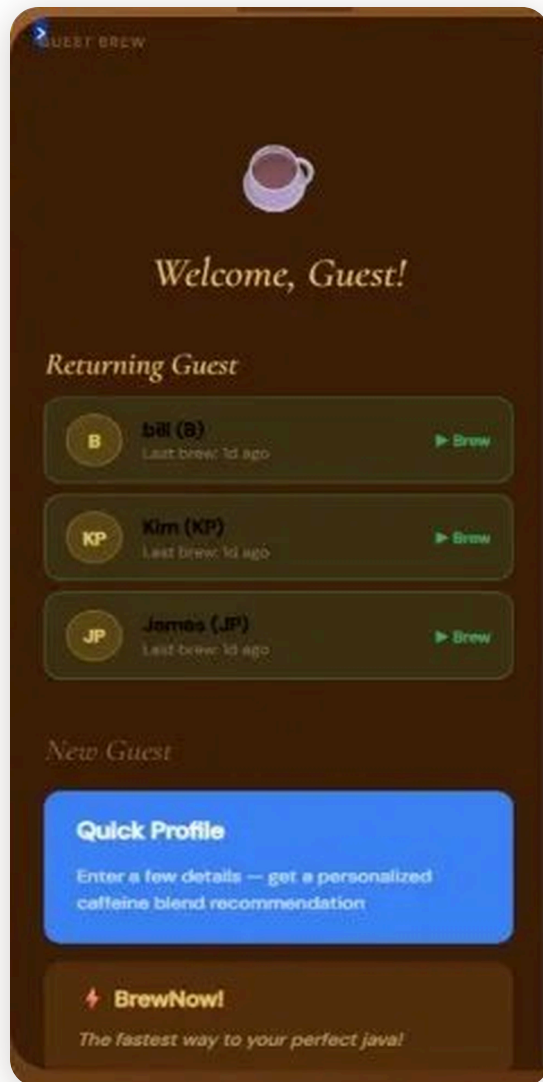
Tap & Hold to Brew
~2 SECONDS

PS6-A — Recommendation & Brew

PS6-A — Recommendation & Brew. A personalized caffeine blend recommendation is generated based on the guest's selected sensitivity level and planned bedtime. *Hold to Brew* confirms and initiates the actual brew cycle. If the guest wishes to change the caffeine content of the brew from the recommendation, *Adjust Manually* is tapped to open the blend adjustment bar.

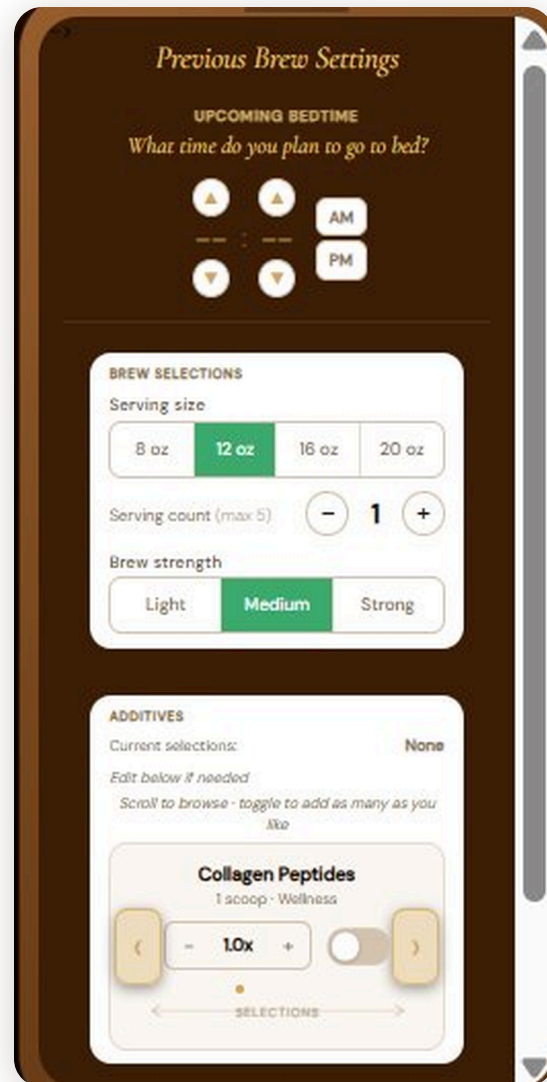
Guest Brew Paths — Returning Guest Path · PS1b · PS2rg · PS6-C1 · PS8

A guest whose Guest Profile was saved as a Returning Guest Profile within the past 72 hours has a profile card on PS1b. Tapping it loads their saved settings directly for confirmation or adjustment.



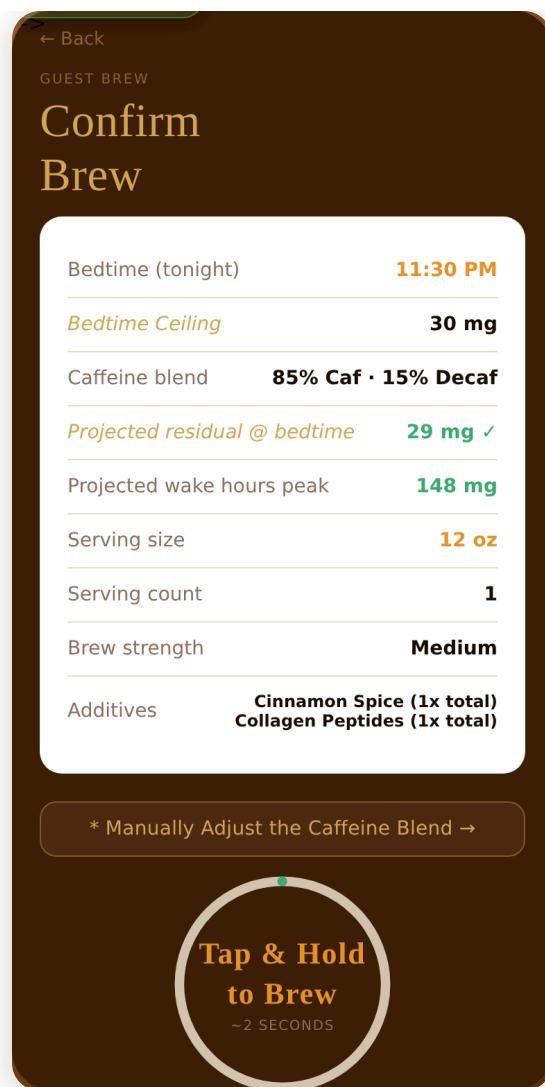
PS1b — Guest Entry

PS1b — Guest Entry. The returning guest's card appears with a direct ► Brew route. Tapping it loads saved settings without re-entering the Quick Profile flow.



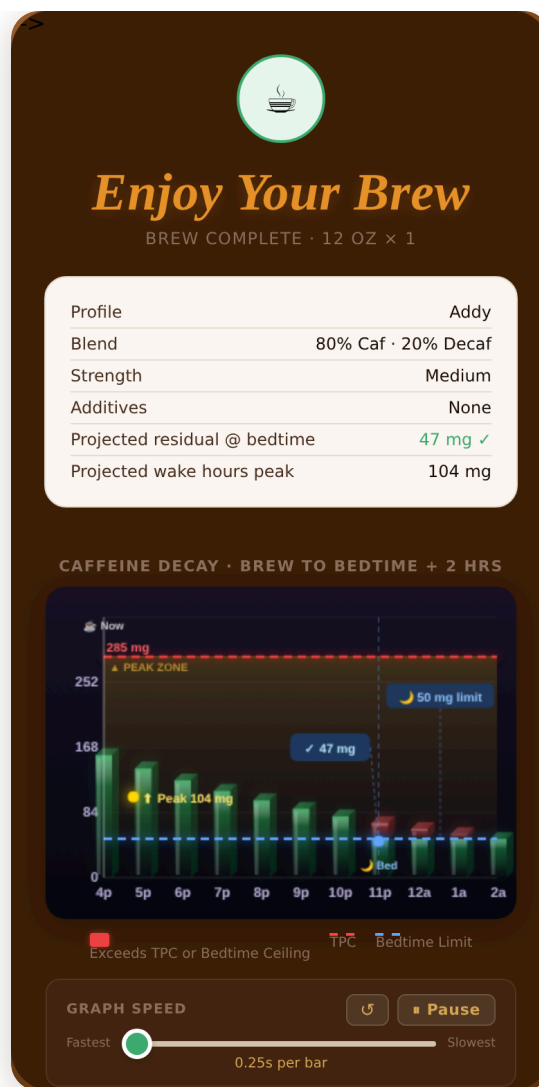
PS2rg — Returning Guest Review

PS2rg — Returning Guest Review. Pre-populated with the guest's last brew settings — size, strength, and additives. The guest confirms or adjusts before continuing.



PS6-C1 — Guest Brew Confirmation

PS6-C1 — Guest Brew Confirmation. The blend recommendation, brew parameters, and selected additives are displayed for final review before *Hold to Brew* is held to initiate the brew cycle. The recommended caffeine blend can be changed by tapping *Adjust Manually*.



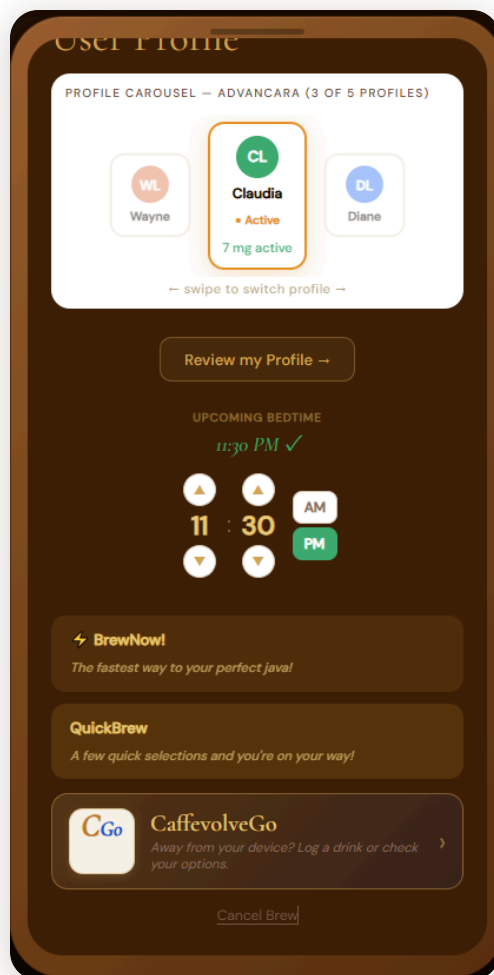
PS8 — Enjoy Your Brew

PS8 — Enjoy Your Brew. The brew complete screen. A session summary card confirms the profile, blend, strength, additives, projected residual caffeine at bedtime, and projected wake-hours peak for the brew just completed. A Caffeine Decay chart animates the projected caffeine curve from brew time through bedtime plus two hours, with reference lines marking the Wake Hours Ceiling and Bedtime Ceiling. The screen auto-advances to the home screen after a short countdown; the user may also tap *All set — Thank you!* to return immediately.

CaffeolveGo

CaffeolveGo is reached from the profile screen (PS1a). Once a bedtime is set for the session, the user taps the *CaffeolveGo* button to open the hub, where the platform's two away-from-home functions are presented as a single branded choice — *External Caffeine Entry (ECE)* to log a drink already consumed, and the *Offsite Advisory System (OAS)* to weigh what to consume next. Both subsections that follow begin here.

PS1a · CaffeolveGo (Hub)



PS1a — Profile Selection

PS1a — Profile Selection. The user selects their profile, confirms the upcoming bedtime, and taps *CaffeolveGo* at the bottom of the screen to open the hub.



CaffeolveGo — Hub

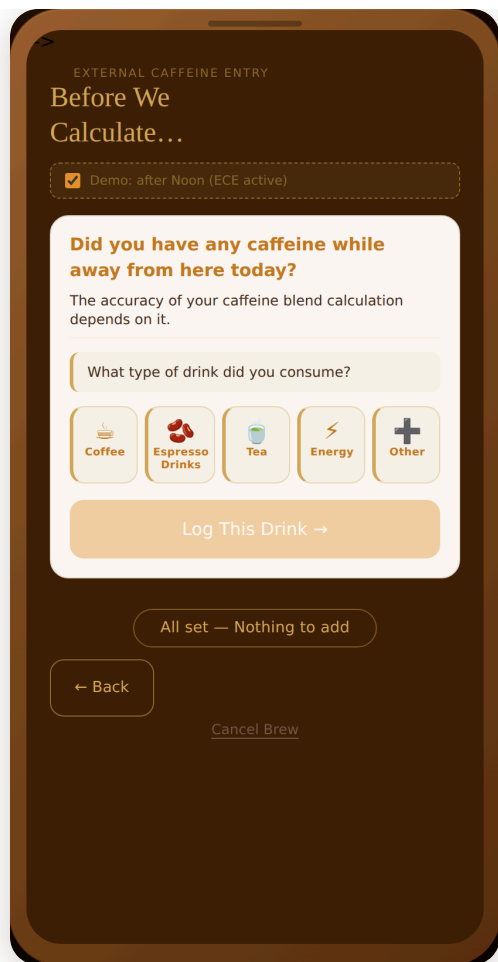
CaffeolveGo — Hub. The hub presents both functions side by side: the user taps *Log a drink* to record an external caffeine event (ECE), or *See my options* to open the Offsite Advisory System (OAS). The two subsections that follow pick up from this screen.

External Caffeine Entry (ECE)

The External Caffeine Entry screens are used to record caffeine activity recorded away from the device — logging a past event to update the residual model and the Offsite Advisory System (OAS).

ECE operates in two modes. **ECEr (Reactive)** is user-initiated at any time — the user opens the ECE option from the CaffevolveGo hub to log an offsite caffeine event on demand. **ECEp (Proactive)** is system-initiated: it fires automatically before a brew or OAS session when the user's last declared bedtime has passed and no device brew or ECE entry has been recorded since that bedtime. Two ECEp windows are defined per wake cycle, each 6 consecutive hours long — Window 1 opens on the first brew or ECE entry after the last declared bedtime has passed, and Window 2 opens 6 hours after Window 1. ECEp fires once per window; answering or skipping suppresses it for the remainder of that window, so it never fires twice for the same gap. Because the daily log is shared in real time between the device and companion app, any caffeine already logged via OAS is present when ECEp opens — the user is asked only to account for what those channels haven't already captured. For guests, the ECEp flag is keyed to the guest session and clears when the session ends.

[Log a Previous Event](#) · ECE



ECE — External Caffeine Entry

ECE — External Caffeine Entry. The user enters drink type, size, servings and approximate time of the external event. The residual model is updated immediately and all blend recommendations in the session reflect the logged intake. Tapping *All set — Nothing to add* skips the entry and returns to the brew selection screen.

ECE — Drink Selection

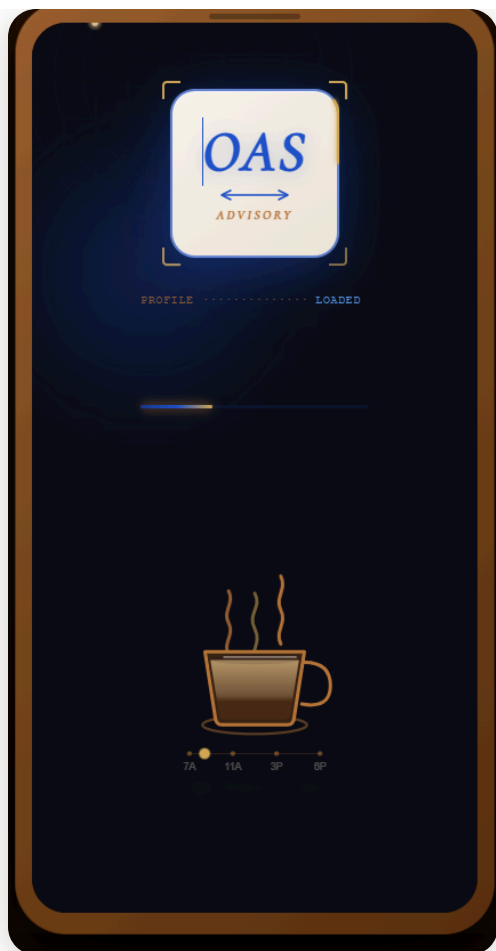
ECE — Drink Selection. After selecting the drink category, the user chooses the specific drink type, sets the number of servings, and selects when the drink was consumed — now, or an earlier time window. Tapping *Log This Drink →* records the entry.

ECE — Session Log

ECE — Session Log. Logged entries appear in the session list with the option to add another drink or continue to brew. Tapping *No — Done, Continue to Brew* closes the ECE session and returns to the brew flow with an updated residual model.

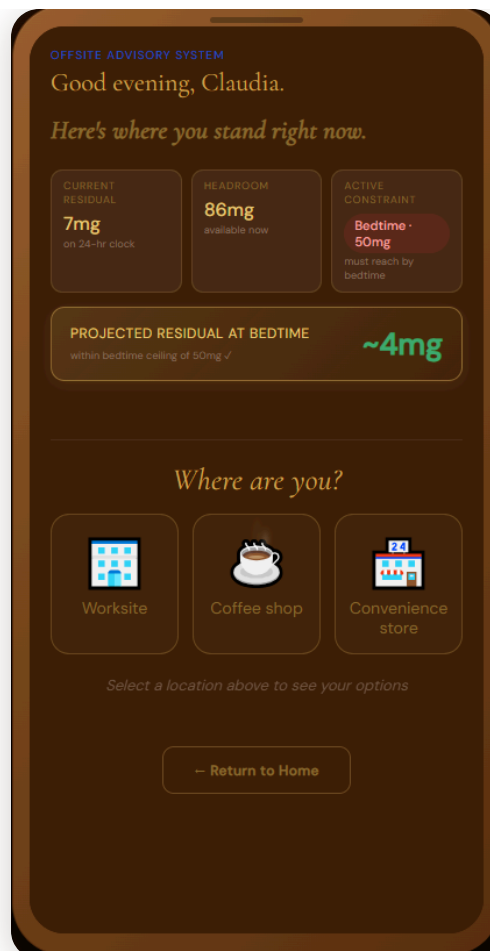
Offsite Advisory System (OAS)

OAS0 (Advisory Launch) · OAS1a (Location Type) · OAS1b (Drink Selection) · OAS2 (Confirm & Log)



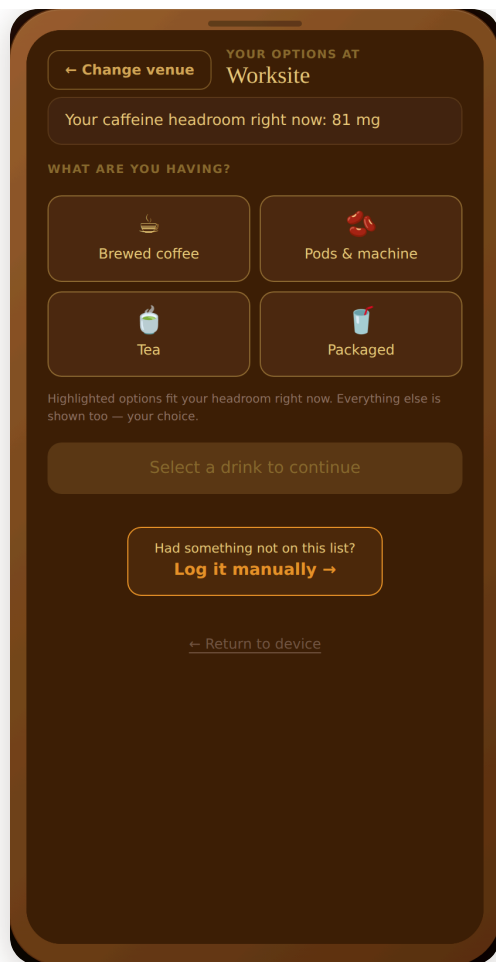
OAS0 — Advisory Launch

OAS0 — Advisory Launch. A brief branded interstitial confirms the profile is loaded and the current standing is being read before the advisory opens.



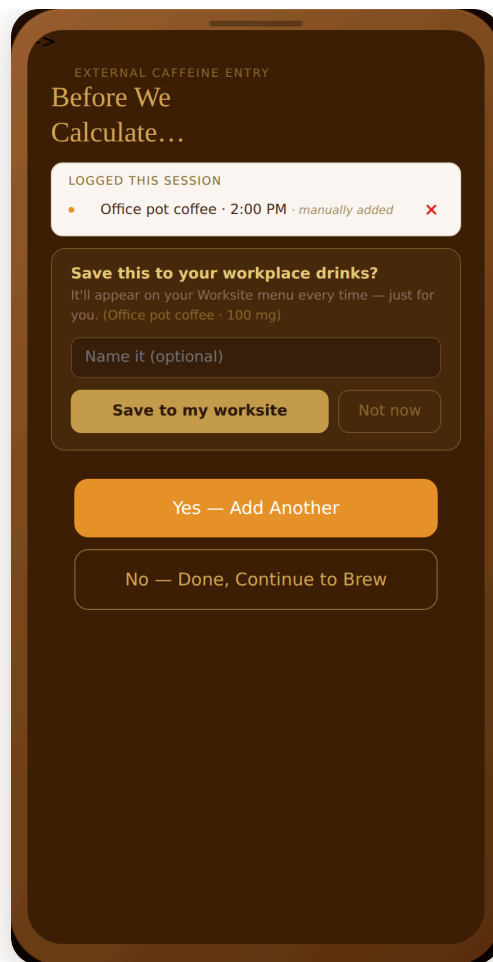
OAS1a — Location Type

OAS1a — Location Type. The OAS opening screen presents the user's current caffeine metrics — residual level, available headroom, and active constraint — alongside a venue selector. The user taps their current location type to load the appropriate drink menu. The venue selector leads with **Worksite** — the workplace setting — followed by the other common venues such as café and convenience store.



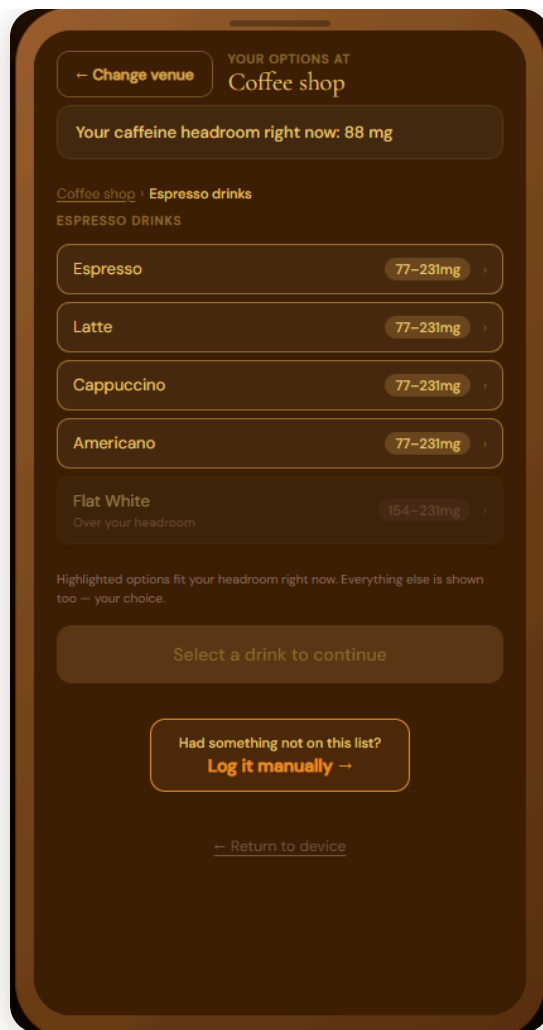
OAS1b — Worksite Venue

Worksite venue. Worksite is the first venue in the OAS selector and covers the drinks a user is most likely to encounter at their workplace, organized into categories — brewed coffee, pods and single-serve machines, tea, and packaged or canned drinks. Because the strength of a shared office pot is not something the user controls, serving volume is the primary lever at this venue: each listed drink is offered on a pour ladder — a full cup, two-thirds, or one-third — so the user can size the drink to the headroom available. The menu presents the full set of options with those that fit the user’s current headroom highlighted; the highlighting is guidance, not a gate, consistent with the platform’s Advisory Philosophy.



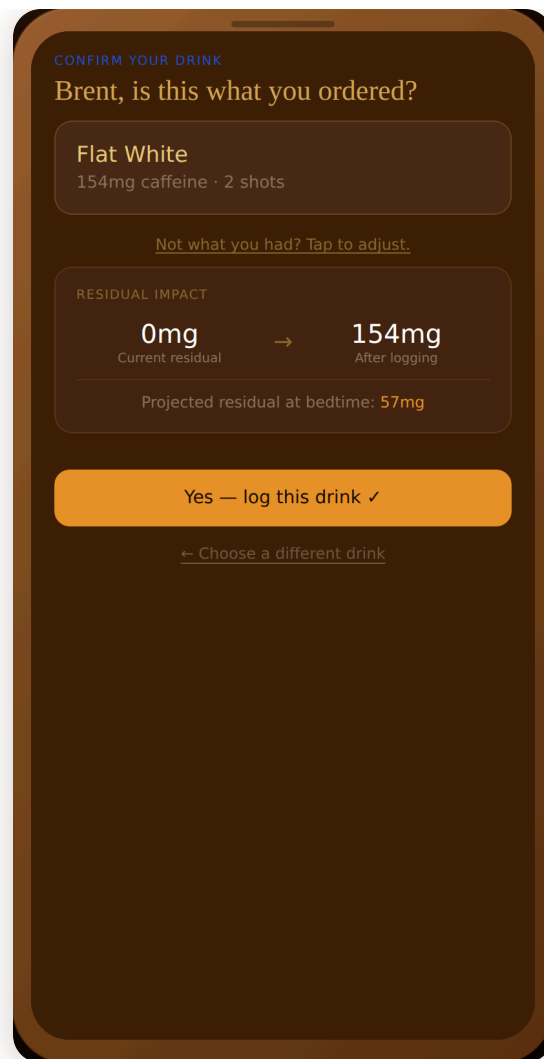
ECE — Save to Workplace Drinks

Saved workplace options. When a user logs a workplace drink that is not on the standard list — using the same Alternative Drink Choice path available elsewhere in the OAS — the platform offers to save that drink to a personal *My workplace drinks* group, with an optional rename. On later visits to the Worksite venue, saved drinks appear alongside the standard options and are highlighted and ranked against the user’s current headroom in the same way. A user manages their saved workplace drinks — renaming or removing them — from the Profile Summary (SU13).



OAS1b — Drink Selection

OAS1b — Drink Selection. The drink menu populates for the selected venue and presents the full set of likely options, with those that fit the user’s current headroom highlighted and ranked first; options that exceed it appear below, de-emphasized rather than hidden. The highlighting is guidance, not a gate — consistent with the platform’s Advisory Philosophy — and a **Log a different drink** path is always available for anything not listed, recorded via the Alternative Drink Choice path. Every selection is written to the shared daily log and reflected immediately in the residual model.



OAS2 — Confirm & Log

OAS2 — Confirm & Log. The selected drink is displayed with its caffeine content and projected impact — current residual before and after logging, and projected bedtime residual. The user taps *Yes — log this drink* to record the event, and return to the Sleep screen.

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